

Connections, Curvature and Symmetry: A Study of G -Invariant Connection and Curvature Geometry on the Frame Bundle over a Symmetric Base Manifold

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ABSTRACT

We investigate G -invariant connections and curvature forms on principal bundles over symmetric base manifolds, using the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$ as a motivating example. We begin by reviewing and carefully constructing the foundational tools required, including differential forms, principal bundles, connections, and curvature, ensuring a rigorous framework for our study. Using this foundation, we analyse the standard $U(1)$ -invariant connection on the Hopf bundle, computing its curvature explicitly and examining its compatibility with the induced action on the frame bundle. This provides a concrete illustration of how horizontal distributions, covariant derivatives, and parallel transport behave under a symmetry group. Building on this example, we formulate a general framework for studying G -invariant connections on the frame bundle of a principal bundle over a symmetric space, employing the foundational tools we will study first to classify such connections and derive explicit curvature expressions. The results elucidate the interplay between symmetry, connection invariance, and curvature, providing a foundation for further exploration of geometric structures on symmetric spaces.

1. Introduction and Motivation

The study of the geometry of principal bundles provide a unifying framework for understanding symmetries in differential geometry. In particular, one is interested how symmetries, encoded by some Lie group G , acts on a space, allowing us to define notions of connections, and hence curvature, on this space. This essay goes the next level up and investigates the invariance properties of connections and curvature forms on the frame bundles of the principal bundles. Our guiding motivation is to study the conditions under which a curvature 1-form and curvature 2-form are invariant under the action of a structure group G .

As a model case we look at the Hopf fibration, the simplest, non-trivial principal $U(1)$ -bundle. This example is computationally explicit and permits direct calculation of the connection and curvature. Our goal here is to formulate definitions for $U(1)$ -invariant connections and curvature on its frame bundle, $\text{Fr}(S^3)$.

Building on this, we turn to the general setting of any principal G -bundle and develop the framework of a G -invariant connection 1-forms, and the corresponding notion of invariance for Ehresmann connections, and show these definitions are equivalent. We then prove how the curvature 2-form satisfies the structure equivalent in the invariance setting, and analyse how parallel transport and the covariant derivative provide a natural explanation for the invariance properties observed.

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Finally, we discuss to what extent these constructions depend on the bundle being principal, or even having a fibre bundle. Specifically, what if a structure group G acts on any manifold M (or any topological space), and how the frame bundle of M can be used to carry over the theory of G -invariant connections and curvature. This leads to a broader perspective on the relationship between symmetry, geometry, and topology.

2. Setting the Scene

2.1. First, Some Algebra

“Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine.”

Atiyah brings home the emphasis on how geometry is more often reliant on multilinear algebra—it is a powerful tool that is metaphorically the nuts and bolts to the theory we develop further on. While intuitively the concepts become harder to grasp, upon reflection they enrich the geometry we do greatly. Therefore sing Atiyah’s quote as inspiration, we begin our study by clarifying some definitions from multilinear algebra including tensors, wedge products, and differential r -forms.

A real-valued, multilinear function T on $V^r = \underbrace{V \times \cdots \times V}_{r \text{ times}}$ is called a **r -tensor**. We define multilinear to mean the function is separately linear in each variable, that is:

$$T(v_1, \dots, \alpha v_i + \beta \tilde{v}_i, \dots, v_r) = \alpha T(v_1, \dots, v_i, \dots, v_r) + \beta T(v_1, \dots, \tilde{v}_i, \dots, v_r)$$

Familiar examples include 1-tensors that are linear functionals on V ; the dot product on $\mathbb{R}^n$ is a 2-tensor; finally the determinant on $\mathbb{R}^n$ is a n -tensor. The collection of all r -tensors is also a vector space, denoted by $\mathfrak{T}(V^*)$.

Let T be a r -tensor, and S be a q -tensor. The **tensor product** of T with S , denoted as $T \otimes S$, is a $r + q$ -tensor and is given by

$$T \otimes S(v_1, \dots, v_r, v_{r+1}, \dots, v_{r+q}) = T(v_1, \dots, v_r) \cdot S(v_{r+1}, \dots, v_{r+q}).$$

The tensor product is *noncommutative*, however it is associative, and distributive over addition. Recall S_r , the group of permutations, and that an element σ of S_r is called *even* if it can be expressed as a product of an even number of transpositions, and *odd* if it is expressible as a product of an odd number of transpositions. Let T be r -tensor, and take any permutation σ in S_r . We define another r -tensor T^σ ,

$$T^\sigma(v_1, \dots, v_r) = T(v_{\sigma(1)}, \dots, v_{\sigma(r)}).$$

We call T an **alternating r -tensor** if it satisfies $T^\sigma = (-1)^\sigma T$ for every σ in S^σ .

For any r -tensor T , we can define a new alternating tensor by

$$\text{Alt}(T) = \frac{1}{r!} \sum_{\sigma \in S^\sigma} (-1)^\sigma T^\sigma.$$

We leave it to the reader to consult (ref differential topology pollack etc page 155) if they wish to see the justification.

As the sums and scalar multiples of alternating functions are also alternating, the collection of alternating p -tensors form a vector subspace $\Lambda^r(V^*)$ of $\mathfrak{T}(V^*)$. For $T \in \Lambda^r(V^*)$ and $S \in \Lambda^q(V^*)$, their **wedge product** is given by

$$\Lambda^{r+q}(V^*) \ni T \wedge S := \text{Alt}(T \otimes S).$$

Note that Alt is a linear operation, therefore the wedge product is distributive over addition and scalar multiplication and a short argument also shows it is associative (ref pollack again page 156/157 and lemma and theorem on that page). Furthermore, for any linear functionals ω and τ on V (so $\omega, \tau \in \Lambda^1(V^*)$), the Alt operator takes the form $\omega \wedge \tau = \frac{1}{2}(\omega \otimes \tau) - \tau \otimes \omega$. This implies the following fundamental properties of the wedge product on 1-forms:

(1) *anticommutativity*:

$$\omega \wedge \tau = -\tau \wedge \omega$$

(2) $\omega \wedge \omega = 0$

A couple more facts to add is that if $r > k$ then $\Lambda^r(V^*) = 0$, and $\Lambda^0(V^*) = \mathbb{R}$. We want to extend wedge product by letting the wedge product of any element in $\mathbb{R}$ with a tensor in $\Lambda^r(V^*)$ be a scalar multiplication. We then get this direct sum

$$\Lambda(V^*) = \bigoplus_{n=0}^k \Lambda^n(V^*).$$

This is called the **exterior algebra** of V^* , with the identity element $1 \in \Lambda^0(V^*)$.

Now we are ready to define differential forms. Let M be a smooth manifold. A **differential r -form** on M is a function $\omega(x) \in \Lambda^r(T_x^*M)$ that assigns an alternating r -tensor ω to each point $x \in M$ on that tangent space of M at the point x . For two r -forms at the same point x on M , adding them yields a new r -form

$$\omega(x) + \theta(x) = (\omega + \theta)(x).$$

Moreover, if ω is a r -form, and τ is a q -form, then $\omega \wedge \tau$ is a $r + q$ form and is given by

$$(\omega \wedge \tau)(x) = \omega(x) \wedge \tau(x).$$

Note that 0-forms are arbitrary real-valued functions on M . If $f : M \rightarrow \mathbb{R}$ is a smooth function, then $df_x : T_x M \rightarrow \mathbb{R}$ is a pointwise linear map. The assignment $x \mapsto df_x$ is a 1-form df on M , called the differential of f . Notice the coordinate functions $x_1, \dots, x_k$ on $\mathbb{R}^k$ yield 1-forms $dx_1, \dots, dx_k$ on $\mathbb{R}^k$. At each v in $\mathbb{R}^k$, we know that $T_v \mathbb{R}^k = \mathbb{R}^k$, the differentials $dx_1, \dots, dx_k$ have the action $dx_i(v)(a_1, \dots, a_k) = a_i$, implying that at each point v on $\mathbb{R}^k$ the linear functions $dx_1(v), \dots, dx_k(v)$ are the standard basis for $(\mathbb{R}^k)^*$. For all strictly increasing sequence indexed by $I = (i_1, \dots, i_r)$, define an r -form on $\mathbb{R}^k$ $dx_I := dx_{i_1} \wedge \dots \wedge dx_{i_r}$. The *Basis theorem* states for a k -dimensional vector space V , where $k \geq 1$, any linearly independent set of exactly k elements that is linearly independent and spans V is automatically a basis for V . Combining this result and $\Lambda^r(T_v^*U) = \Lambda^r((\mathbb{R}^k)^*)$, for any open subset U of $\mathbb{R}^k$, we have the result

PROPOSITION 2.1. *Every r -form on an open subset U of $\mathbb{R}^k$ can be uniquely expressed as a sum*

$$\sum_I f_I dx_I$$

over an increasing indexing set, and functions f_I on U .

This brings us to the end of everything we need from multilinear algebra.

Basic notions and definitions from manifolds will be assumed here. Let M and N be smooth manifolds, and $f : M \rightarrow N$ a smooth map between them. The collection of maps

$$f_* : TM \rightarrow TN$$

is called the **push-forward**. They are pointwise linear maps, $f_* : T_x M \rightarrow T_{f(x)} N$ for any point x on M and defined via the following notion. Take a curve $\gamma : I \subset \mathbb{R} \rightarrow M$, $t \mapsto \gamma(t)$ through

the point x on M , where $\gamma(0) = x$ and $\gamma'(0) = X$. That is, X represents the velocity of the curve γ and is the tangent space of M at x . Thus, $f_*(X)$ in $T_{f(x)}N$ is the velocity vector at the point $f(x)$ of the curve $f \circ \gamma : I \subset \mathbb{R} \rightarrow N$, $t \mapsto f(\gamma(t))$, namely $f_*(\gamma'(0)) = (f \circ \gamma)'(0)$.

Let $g : N \rightarrow K$, be another smooth map between manifolds. The **chain rule** is then $(g \circ f)_* = g_* \circ f_*$. Note that the push-forward can also be called the *differential*, and these names will be used interchangeably throughout this text.

Dual to the push-forward is the **pull-back**. Given a r -form ω on N , we can *pull back* ω and define $f^*\omega$ as a r -form on M . The definition is accompanied by the map

$$f^* : T_{f(x)}^*N \rightarrow T_x^*M,$$

and observe the pull-back is only defined on the image of f . For example, if ω is a 1-form on N , then $(f^*\omega)(X) = \omega(f_*X) = \omega(dfX)$. Finally, we have the relation $(g \circ f)^* = f^* \circ g^*$.

2.2. Lie Technology

One of our focuses will be on *continuous symmetry groups*- groups of transformations that act on an object and leave this object, or perhaps another one, invariant. These groups are conceptualised as **Lie groups**: groups that are at the same time a manifold such that both the following maps

$$\begin{aligned} G \times G &\rightarrow G, & (g, h) &\mapsto g \cdot h \\ G &\rightarrow G, & g &\mapsto g^{-1} \end{aligned}$$

are smooth, for any elements g, h in a group G . Lie groups satisfy all the group axioms since they are in their own right a group in the algebraic sense, just equipped with the additional smooth manifold structure. They are a continuous group that can be parametrised locally by coordinates with the group operations of multiplication and inverse defined by smooth maps in these coordinates.

Let L be a finite-dimension vector space and define the **Lie bracket** to be the map

$$[\cdot, \cdot] : L \times L \rightarrow L$$

satisfying the following properties:

- (1) it is a bilinear map;

$$[\alpha t + \beta v, \beta w] = \alpha[t, w] + \beta[v, w]$$

$$[v, \alpha t + \beta w] = \alpha[v, t] + \beta[v, w]$$

- (2) it is antisymmetric; for every v, w in L we have

$$[v, w] = -[w, v]$$

- (3) it satisfies the *Jacobi identity*; for every t, w, v in L we have

$$[t, [v, w]] + [w, [t, v]] + [v, [w, t]] = 0$$

We call the pair $(L, [\cdot, \cdot])$ a **Lie algebra**. A particular example of Lie bracket we will use is the *commutator*, which is defined by

$$[A, B] = AB - BA$$

for every A, B in L . Take M to be a smooth manifold, and $\mathfrak{X}(M)$ as the real vector space of smooth vector fields on M .

For any element g in G , we define the following diffeomorphisms

$$L_g : G \rightarrow G, \quad h \mapsto g \cdot h$$

and

$$R_g : G \rightarrow G, \quad h \mapsto h \cdot g$$

as the **left** and **right translations** respectively.

From manifold theory, the set of smooth vector fields, $\mathfrak{X}(G)$ on M equipped with *commutator* forms a *Lie algebra*. A vector field $X \in \mathfrak{X}(G)$ is said to be **left-invariant** if it satisfies

$$(L_g)_*X = X \quad \text{or} \quad v(g) = (L_g)_*v(e)$$

for any element g in G , and the identity element e of G . Taking the Lie bracket of two left-invariant vector fields yields a left-invariant vector field.

The Lie algebra $\mathfrak{g}$ of a Lie group G is characterised by the vector space of left-invariant vector fields. This vector space of left-invariant vector fields is uniquely determined by its value at the identity e , hence we get $\mathfrak{g} \cong T_e G$. A $\mathfrak{g}$ -valued 1-form is a differential 1-form that assigns an element of the Lie algebra to each point on the manifold, instead of a real number.

The **left-invariant Maurer-Cartan form** is the $\mathfrak{g}$ -valued 1-form on G given by

$$(L_{g^{-1}})_* : T_g G \rightarrow T_e G \quad (\text{or can just put } \mathfrak{g} \text{ instead of } T_e G.)$$

It is also the natural identification between $T_e G$ and $\mathfrak{g}$ because, if X is a left-invariant vector field, then $(L_{g^{-1}})_*(X) = X(e)$. Define $(L_{g^{-1}})_*$ to be the $\mathfrak{g}$ -valued Maurer-Cartan 1-form, $(L_{g^{-1}})_* = \theta_g$; then it satisfies the **structure equation**

$$d\theta = -\frac{1}{2}[\theta, \theta].$$

In particular, let $\{e_i\}_{i \in I}$ be a basis for $\mathfrak{g}$, and $[e_i, e_j] = \sum_k c_k^{i,j} e_k$, where the $c_k^{i,j}$ are structure constants, and θ_i be $\mathfrak{g}$ -valued Maurer-Cartan 1-form on G , equal to the dual basis of $\mathfrak{g}^*$ at the identity. Then

$$d\theta_k = -\frac{1}{2} \sum_{i,j} c_k^{i,j} \theta_i \wedge \theta_j,$$

thus the right-hand side is $-\frac{1}{2}[\theta, \theta]$. For a matrix group we write $\theta = g^{-1}dg$.

Every element g in G defines the **adjoint map**, $L_g \circ R_g^{-1} : G \rightarrow G$ such that

$$L_g \circ R_g^{-1}(h) = ghg^{-1}.$$

It preserves the identity, consequently its derivatives at the identity gives the **adjoint representation**

$$(L_g \circ R_g^{-1})_* = \text{Ad}_g : \mathfrak{g} \rightarrow \mathfrak{g},$$

and is defined by

$$\text{ad}_g X = \frac{d}{dt}(ge^{tX}g^{-1})\Big|_{t=0}.$$

It is a linear representation of the group on the Lie algebra. For a matrix group G we have $\text{ad}_g(X) = gXg^{-1}$.

Notice that $R_g^* \theta = \text{ad}_{g^{-1}} \circ \theta$. (add justification)

2.3. Fibre Bundles

Before moving on to our main setting for this paper, we introduce the more general concept of *fibre bundles*. Fix the following map $\pi : E \rightarrow M$ to be a smooth surjection between smooth manifolds.

For any point $p \in M$, the **fibre** over r is the following non-empty subset $E_p := \pi^{-1}(p)$. Let U be a subset of M , and set $E_U = \pi^{-1}(U)$ to be the union of the fibres E_p . Suppose F is another manifold. If for every point $p \in M$ there is an open neighbourhood $U \subset M$ of r , such

that there exists a diffeomorphism $\phi_U : E_U \rightarrow U \times F$, and a projection $\text{pr}_U : U \times F \rightarrow U$ with $\text{pr}_U \circ \phi_U = \pi$, therefore the diagram below commutes,

$$\begin{array}{ccc} E_U & \xrightarrow{\phi_U} & U \times F \\ & \searrow \pi & \swarrow \text{pr}_U \\ & U & \end{array}$$

If the above holds, we call $(E, \pi, M; F)$ a **fibre bundle**.

We will use the following terminology: E is called the *total space*; M is the *base manifold*; F denotes the *general fibre*; π is the *projection* from E onto M ; finally (U, ϕ_U) is a *bundle chart*. Fibre bundles just generalise the product $M \times F = E$, but only almost look like the product of two topological spaces, since the space as a whole can be twisted in any way. The fibres on a fibre bundle are still embedded submanifolds and diffeomorphic to each other by *Ehresmann's Theorem*. This twisted product element alludes to the fact our focus is on the local product, rather than global.

To understand this further, we introduce another notion: a **local section** is a differentiable map $s : U \rightarrow E$, satisfying

$$\pi \circ s = \text{Id}_U.$$

A **global section** is a differentiable map $s : M \rightarrow E$ such that

$$\pi \circ s = \text{Id}_M.$$

Now, we are finally ready to paint the scenery onto our fibre bundles using the tools we have so far. A **principal fibre bundle** is characterised by a total space P , and a Lie group G that acts *freely* on P on the right

$$P \times G \rightarrow P$$

$$(p, g) \mapsto p \cdot g.$$

A free action is one where every point on P is fixed by only the identity element of G - more technically, the stabiliser of every point on P has no non-trivial stabilisers, $G_p = \{e\}$. We will use without proof that the $M = P/G$ is the base manifold and is also the space of orbits

$$M = P/G := \{\mathcal{O}_p : p \in P\}$$

where $\mathcal{O}_p = \{p \cdot g : g \in G\} \subseteq P$.

We stick to the notation used for general fibre bundles. That is, $\pi : P \rightarrow M$ is a smooth surjective map. Once again, we subject this data to local triviality- for any open subset U of M from a collection of open subsets $\{U_\alpha\}$ that cover M , and G -equivariant diffeomorphisms, $\phi : P_\alpha \rightarrow U_\alpha \times G$, such that we have the commutative diagram

$$\begin{array}{ccc} P_U & \xrightarrow{\phi} & U \times G \\ & \searrow \pi & \swarrow \text{pr}_U \\ & U & \end{array}$$

G -equivariance means given a fibrewise diffeomorphic G -equivariant map

$$g_U : P_U \rightarrow G,$$

$g_U(p \cdot g) = g_U(p)g$. Putting all this together we have $\phi(p) = (\pi(p), g_U(p))$. We will abbreviate all this information by calling $(P, \pi, M : G)$ a principal G -bundle.

A principal G -bundle is **trivial** if there exists a diffeomorphism $\Phi : P \rightarrow M$ with $\Phi(p) = (\pi(p), \chi(p))$, and $\chi(p \cdot g) = \chi(p) \cdot g$. The definition of a section remains unchanged- it is still a map $s : M \rightarrow P$ such that $\pi \circ s = \text{Id}_M$. Global sections can be rare, but we have this proposition:

PROPOSITION 2.2. *A principal G -bundle admits a global section if and only if it is trivial.*

Proof. $\Rightarrow$ Suppose $s : M \rightarrow P$ is a global section, $\pi \circ s = \text{id}_M$. Define

$$\Phi : P \rightarrow M \times G$$

$$p \mapsto (\pi(p), \chi(p))$$

where $\chi(p) \in G$ is the unique element such that $p = s(\pi(p)) \cdot \chi(p)$.

Define

$$\Phi : M \times G \rightarrow P, \quad \Phi(x, g) = s(x) \cdot g,$$

where $\cdot$ denotes the right G -action on P .

- *Well-defined:* Such $\chi(p)$ exists and is unique because the G -action is free and transitive on each fibre.
- *Inverse map:* Define

$$\Psi : M \times G \rightarrow P, \quad \Psi(x, g) = s(x) \cdot g.$$

Then $\Phi(\Psi(x, g)) = (x, g)$ and $\Psi(\Phi(p)) = p$.

- *Continuity/smoothness:* In a local trivialization $P|_U \cong U \times G$, the section has the form $s(x) = (x, k(x))$. For $p = (x, h)$ we get $\chi(p) = k(x)^{-1}h$, which is continuous/smooth in (x, h) . Hence Φ and Ψ are continuous/smooth globally.
- *Equivariance:* For $h \in G$,

$$\Phi(p \cdot h) = (\pi(p), \chi(p \cdot h)) = (\pi(p), \chi(p)h) = \Phi(p) \cdot h.$$

Thus Φ is an isomorphism of principal bundles, so P is trivial.

$\Leftarrow$ On the other hand, given a diffeomorphism $\Phi : P \cong M \times G \rightarrow M \times G$, for a point x on M , we can define a section by

$$s(m) = \Phi^{-1}(m, e)$$

Then we have $\pi \circ s = \pi \circ \Phi^{-1} : M \times G \rightarrow P$, and

$$\begin{aligned} \pi \circ s(x) &= \pi \circ \Phi^{-1}(x, e) \\ &= \text{Id}_M \end{aligned}$$

for any point x on M , and the identity element e of G . □

Indeed local sections do exist because P is locally trivial. As a result, we have local sections $s_U : U \rightarrow P_U$ associated to the local trivialisation of the principal bundle. At every point x in U , add identity e of G we have $\phi(s_U(x)) = (x, e)$, implying $g_U \circ s : U \rightarrow G$ is a constant function that sends every point to the identity.

On the other hand, we can identify a fibre over x with G using a local section such that given any $p \in \pi^{-1}(x)$, there exists a unique element $g_U(p) \in G$ with $p = s_U(x)g_U(p)$.

EXAMPLE 1.

The Möbius band $\mathbb{M}$ is a classic example of a non-trivial principal bundle due to the twist that prevents it from being globally a product space $S^1 \times \mathbb{R}$. In particular, the Möbius band can be

explicit defined by

$$\mathbb{M} = \frac{[0, 1] \times \mathbb{R}}{(0, t) \sim (1, -t)}.$$

The projection

$$\pi : S^1 \rightarrow S^1, \quad \pi(z) = z^2,$$

is a real line bundle with fibre $\mathbb{R}$. The fibre

$$\pi^{-1}(z^2) = \{\pm z\}$$

consists of two points.

The structure group G of this bundle is $\mathbb{Z}_2 = \{+1, -1\}$, acting on $\mathbb{R}$ by multiplication. The principal bundle is obtained by restricting to the “unit vectors” in each fibre:

$$P = \{(p, q) \in \mathbb{M} : |q| = 1\}.$$

Then P is a principal $\mathbb{Z}_2$ -bundle over S^1 , with the $\mathbb{Z}_2$ -action given by $(p, q) \cdot g = (p, gq)$ for $g \in \mathbb{Z}_2$.

A global section choosing a square root continuously around the circle would mean finding a continuous function

$$z^{1/2}$$

on the unit circle. Such a function does not exist due to the presence of a branch cut in the complex plane. Topologically, if the bundle were trivial, the total space would be two disjoint copies of the circle. However, the boundary of the Möbius band is connected, demonstrating the non-triviality of the bundle.

Locally, the Möbius band can be covered by two open sets U_1 and U_2 , each trivializing the bundle. The intersection $U_1 \cap U_2$ consists of two disconnected intervals $V_1 \sqcup V_2$. The transition functions

$$g_i : V_i \rightarrow \mathbb{Z}_2,$$

being constant on connected components, satisfy the cocycle condition trivially. Since the transition functions differ by the non-trivial element of $\mathbb{Z}_2$ in one of the overlaps, the bundle exhibits the characteristic twist of the Möbius strip and thus is non-trivial.

Consequently, we can generalise the idea of trivialising the bundles on non-empty overlaps. Suppose $\pi : P \rightarrow M$ is a principal G -bundle and $\{U_\alpha\}$ is an open cover of M over which P is trivial. For each α , a trivialisation

$$\phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times G$$

is chosen. On the overlaps $U_{\alpha\beta} := U_\alpha \cap U_\beta$ we have two such trivialisations, and therefore two descriptions of each point in the fibre. This situation can be expressed by the diagram

$$\begin{array}{ccccc} U_{\alpha\beta} \times G & \xleftarrow{\phi_\beta} & \pi^{-1}(U_{\alpha\beta}) & \xrightarrow{\phi_\alpha} & U_{\alpha\beta} \times G \\ & \searrow \text{pr}_1 & \downarrow \pi & \swarrow \text{pr}_1 & \\ & & U_{\alpha\beta} & & \end{array}$$

Let $x \in U_{\alpha\beta}$ and $p \in \pi^{-1}(x)$. Then

$$\phi_\alpha(p) = (x, g_\alpha(p)) \quad \text{and} \quad \phi_\beta(p) = (x, g_\beta(p)).$$

Since both describe the same point p , there exists an element $\tilde{g}_{\alpha\beta}(p) \in G$ such that

$$g_\alpha(p) = \tilde{g}_{\alpha\beta}(p) g_\beta(p).$$

Equivalently,

$$\tilde{g}_{\alpha\beta}(p) = g_\alpha(p)g_\beta(p)^{-1}. \quad (1)$$

The quantity $\tilde{g}_{\alpha\beta}(p)$ is constant along each fibre. Indeed, if $p \in \pi^{-1}(m)$ and $h \in G$, then

$$\begin{aligned} \tilde{g}_{\alpha\beta}(ph) &= g_\alpha(ph)g_\beta(ph)^{-1} \\ &= g_\alpha(p)h(g_\beta(p)h)^{-1} \\ &= g_\alpha(p)g_\beta(p)^{-1} \\ &= \tilde{g}_{\alpha\beta}(p), \end{aligned}$$

using the equivariance of the local representatives g_α, g_β . Thus $\tilde{g}_{\alpha\beta}(p)$ depends only on the base point $x = \pi(p)$, and we can define

$$g_{\alpha\beta} : U_{\alpha\beta} \rightarrow G, \quad g_{\alpha\beta}(m) := \tilde{g}_{\alpha\beta}(p).$$

From equation (1) it follows that these transition functions satisfy:

1. On double overlaps $U_{\alpha\beta}$,

$$g_{\alpha\beta}(x)g_{\beta\alpha}(x) = e, \quad \text{for all } x \in U_{\alpha\beta}. \quad (2)$$

2. On triple overlaps $U_{\alpha\beta\gamma} := U_\alpha \cap U_\beta \cap U_\gamma$,

$$g_{\alpha\beta}(x)g_{\beta\gamma}(x)g_{\gamma\alpha}(x) = e, \quad \text{for all } x \in U_{\alpha\beta\gamma}. \quad (3)$$

These are called the **cocycle conditions**.

Each trivialisation ψ_α determines a canonical local section

$$s_\alpha : U_\alpha \rightarrow P, \quad s_\alpha(m) = \psi_\alpha^{-1}(m, e).$$

On overlaps $U_{\alpha\beta}$, these are related by

$$s_\beta(m) = s_\alpha(m)g_{\alpha\beta}(m).$$

Conversely, given an open cover $\{U_\alpha\}$ of M together with transition functions $\{g_{\alpha\beta}\}$ satisfying the cocycle conditions (2)–(3), one can reconstruct a principal bundle by setting

$$P := \bigsqcup_{\alpha} (U_\alpha \times G) / \sim,$$

where the equivalence relation is

$$(x, g) \sim (x, g_{\alpha\beta}(x)g) \quad \text{for all } x \in U_{\alpha\beta}, g \in G.$$

The projection $\pi : P \rightarrow M$ is induced by projection to the first factor, and the right G -action on P is induced by multiplication on the second factor. The cocycle conditions ensure that this action is well-defined and associative.

2.4. The Frame Bundle as a Principal Bundle

One of the central insights of this framework is that the seemingly abstract concept of a principal bundle is not a construction foreign to geometry, but something that is already present in the very structure of a manifold. The canonical example of this is the *frame bundle*, which serves as the prototype of a principal bundle in differential geometry. To understand this properly, it is helpful to retrace the argument showing why the frame bundle of a smooth manifold is indeed a principal $GL(n, \mathbb{R})$ -bundle.

We begin with a smooth n -dimensional manifold M . At every point $x \in M$, we have the tangent space $T_x M$, an n -dimensional real vector space. Choosing a frame at x means selecting an ordered basis of this tangent space, i.e. a tuple $(e_1, \dots, e_n)$ such that any tangent vector can be written uniquely as a linear combination of the e_i . Intuitively, a frame provides us with a

“coordinate system” internal to the tangent space at that point: once a frame is fixed, we can describe tangent vectors at x using their coordinate components with respect to that basis. Now, one does not want to choose a single frame, but rather consider all possible frames at every point of the manifold. This leads us to define the **frame bundle** $\text{Fr}(M)$. By definition, the fibre over $x \in M$, denoted $\text{Fr}_x(M)$, is the set of all ordered bases of $T_x M$. The total space of the frame bundle is then the disjoint union of these fibres,

$$\text{Fr}(M) = \bigsqcup_{x \in M} \text{Fr}_x(M),$$

together with the natural projection map

$$\pi : \text{Fr}(M) \rightarrow M, \quad \pi(e_1, \dots, e_n) = x.$$

Thus, the frame bundle is built by “stacking together” all the possible bases of tangent spaces across the manifold.

At this point, we see the role of the structure group G . Each frame can be transformed into another frame at the same point by applying an invertible linear transformation. This is precisely the action of the general linear group $GL(n, \mathbb{R})$. Explicitly, given a frame $(e_1, \dots, e_n)$ at x and a matrix $B = (b_{ij}) \in GL(n, \mathbb{R})$, we obtain a new frame

$$(e_1, \dots, e_n) \cdot B = \left(\sum_{i=1}^n e_i b_{i1}, \sum_{i=1}^n e_i b_{i2}, \dots, \sum_{i=1}^n e_i b_{in} \right).$$

In other words, we take linear combinations of the old basis vectors with coefficients given by the columns of the matrix B , yielding a new ordered basis. This group action is completely natural: it is nothing more than the familiar operation of changing bases in a vector space.

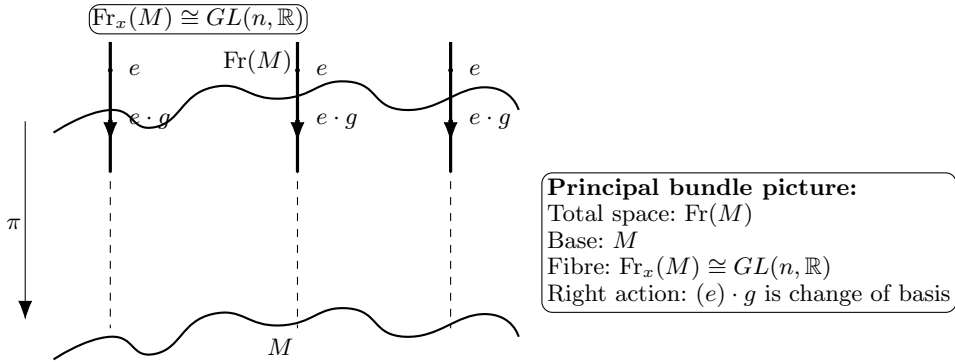


FIGURE 1. The frame bundle $\text{Fr}(M)$ over M : each fibre $\text{Fr}_x(M)$ is a principal homogeneous space for $GL(n, \mathbb{R})$.

Now we verify the essential properties that make this a principal bundle action. First, the action is free: if applying a matrix B leaves the frame unchanged, then B must be the identity. Second, the action is transitive on each fibre: given any two frames at the same point x , there exists a unique $B \in GL(n, \mathbb{R})$ that maps one frame to the other. This uniqueness follows from the fact that a basis determines a linear isomorphism to $\mathbb{R}^n$, so comparing two bases gives a unique change-of-basis matrix. Therefore, each fibre $\text{Fr}_x(M)$ is not just an arbitrary set of frames, but rather a principal homogeneous space for the group $GL(n, \mathbb{R})$.

What remains is to show that this bundle is not just a set-theoretic construction but a smooth manifold in its own right, and that the group action is smooth. Here local coordinates come to the rescue. Suppose $(U, x^1, \dots, x^n)$ is a coordinate chart on M . Then at any point $x \in U$, the

$$\begin{array}{ccc}
 \pi^{-1}(U) & \xrightarrow{\Phi} & U \times GL(n, \mathbb{R}) \\
 \pi \downarrow & & \downarrow \text{pr}_1 \\
 U & \xlongequal{\quad} & U
 \end{array}$$

Here $U \subset M$ is a coordinate chart and Φ is the smooth trivialization sending a frame $e = (e_1, \dots, e_n)$ at $x \in U$ to (x, B) , where $B \in GL(n, \mathbb{R})$ is the change-of-basis matrix from the coordinate frame $(\frac{\partial}{\partial x^1}|_x, \dots, \frac{\partial}{\partial x^n}|_x)$ to e . The right action corresponds to $(x, B) \cdot g = (x, Bg)$.

FIGURE 2. Local trivialization of the frame bundle over a chart U .

coordinate vector fields

$$\left\{ \frac{\partial}{\partial x^1} \Big|_x, \dots, \frac{\partial}{\partial x^n} \Big|_x \right\}$$

form a canonical frame of $T_x M$. Any other frame at x can be expressed uniquely as a linear transformation of this coordinate frame. Thus, choosing a frame amounts to choosing an invertible matrix $B \in GL(n, \mathbb{R})$.

This gives us a concrete parametrization:

$$\pi^{-1}(U) \cong U \times GL(n, \mathbb{R}),$$

where a frame at $x \in U$ corresponds to the pair (x, B) , with A recording the change of basis from the coordinate frame to the chosen frame. This identification is smooth, and the transition maps between overlapping coordinate charts are given by multiplication by smooth Jacobian matrices, which are themselves elements of $GL(n, \mathbb{R})$. Therefore, the total space $\text{Fr}(M)$ inherits the structure of a smooth manifold.

Finally, under this local trivialization, the group action of $GL(n, \mathbb{R})$ becomes nothing more than the standard right action on the second factor:

$$(x, B) \cdot g = (x, Bg).$$

This demonstrates both the smoothness of the action and its compatibility with the bundle projection.

To summarize, the frame bundle $\text{Fr}(M)$ has fibres consisting of frames of tangent spaces, these fibres are principal homogeneous spaces under $GL(n, \mathbb{R})$, the total space is a smooth manifold locally trivialized as $U \times GL(n, \mathbb{R})$, and the right action of the group is free, transitive, and smooth. By definition, this makes $\text{Fr}(M)$ into a principal $GL(n, \mathbb{R})$ -bundle.

This result reveals that the very geometry of a manifold naturally gives rise to a principal bundle structure. The frame bundle is not an added decoration but a canonical object, embodying the local symmetries of linear algebra within the geometry of the manifold.

2.5. Stay Connected!

At this point we have a base space, with fibres that look like copies of a group G hovering freely above each point. But now we want to probe deeper: how can we compare these fibres at different points of the base? The natural answer is to *connect* them. The mathematical tool that allows us to do so is, fittingly, called a *connection* (who says mathematicians lack creativity). There will be two perspectives of connections below: the first will be more geometric, while the second is algebraic. However, Atiyah's words still hang over us, and so we will see a theorem that bridges the two viewpoints. Before jumping in, we give a definition.

Let $(P, \pi, M : G)$ be a principal G -bundle, and as before we have x , a point on M , and $p = \pi^{-1}(x)$ on P . The **vertical tangent space**, V_p , is a vector subspace of the tangent space of P at p , and contains all the vectors that are tangent to the fibre at p . Paraphrasing this, given the differential of π at p , $\pi_* : T_p P \rightarrow T_x M$, the vertical tangent space is $V_p = \ker(\pi_*)$. We say a

vector field $X \in \mathfrak{X}(P)$ is **vertical** if it is an element in the vertical subspace V_p for every $p \in P$ - $X_p \in V_p$ for any p . One can check that the Lie bracket of two vertical vector fields is again vertical, so the vertical subspaces together form a G -invariant distribution $V \subset TP$. Indeed, if $R_g : P \rightarrow P$ denotes the right action of $g \in G$, then

$$(R_g)_* V_p = V_{pg}.$$

In general, there is no canonical complement to V_p inside $T_p P$. A connection provides such a complement by specifying a consistent choice of horizontal directions.

An **Ehresmann connection** on P is a smooth assignment of subspaces

$$H_p \subset T_p P$$

such that

$$T_p P = V_p \oplus H_p,$$

and the distribution $H := \{H_p\}_{p \in P}$ is G -invariant:

$$(R_g)_* H_p = H_{pg}, \quad \text{for all } g \in G.$$

In other words, an Ehresmann connection splits the tangent space at each point into vertical and horizontal components, in a manner compatible with the group action.

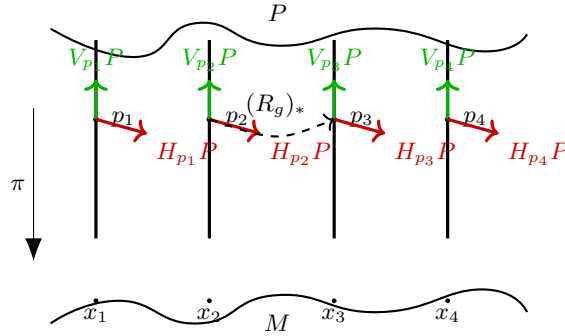


FIGURE 3. *Ehresmann connection on a principal bundle: each fibre $\pi^{-1}(x_i)$ contains a point p_i , where the tangent space splits as $T_{p_i} P = V_{p_i} P \oplus H_{p_i} P$. Vertical spaces (green) are tangent to fibres, while horizontal spaces (red) provide a complementary distribution, preserved by the right action.*

The action of G on P gives a natural way to associate elements of the Lie algebra $\mathfrak{g}$ with vertical vector fields. For $X \in \mathfrak{g}$, the **fundamental vector field** $\tilde{X}$ is defined at $p \in P$ by

$$\tilde{X}_p = \left. \frac{d}{dt} (pe^{tX}) \right|_{t=0}.$$

Since $\pi(pe^{tX}) = \pi(p)$ for all t , we have

$$\pi_* \tilde{X}_p = \left. \frac{d}{dt} \pi(pe^{tX}) \right|_{t=0} = 0,$$

so $\tilde{X}$ is always vertical.

In fact, because the action of G on P is free, the map

$$\sigma_p : \mathfrak{g} \rightarrow V_p$$

is a linear isomorphism for every $p \in P$. Thus, the vertical space at p can be identified with $\mathfrak{g}$ itself.

LEMMA 2.3. For any $g \in G$ and $X \in \mathfrak{g}$, we have

$$(R_g)_*(\tilde{X}) = \sigma(\mathrm{ad}_{g^{-1}} X).$$

This identity expresses how fundamental vector fields transform under the right action of the group.

Proof. By evaluating at $p \in P$,

$$\begin{aligned} (R_g)_*\tilde{X}_p &= \left. \frac{d}{dt} \right|_{t=0} R_g(pe^{tX}) \\ &= \left. \frac{d}{dt} \right|_{t=0} (pe^{tX}g) \\ &= \left. \frac{d}{dt} \right|_{t=0} (pg(g^{-1}e^{tX}g)) \\ &= \left. \frac{d}{dt} \right|_{t=0} (pge^{t\mathrm{ad}_{g^{-1}} X}), \end{aligned}$$

which shows the push-forward equals the fundamental vector field at $p \cdot g$ corresponding to $\mathrm{ad}_{g^{-1}} X$,

$$= \sigma_{pg}(\mathrm{ad}_{g^{-1}} X).$$

□

LEMMA 2.4. Let $g : P_U \rightarrow G$ be the local trivialisation map. Then

$$g_*\tilde{X}_p = (L_{g(p)})_*X.$$

Proof. Using definitions and the equivariance of g we have:

$$\begin{aligned} g_*\tilde{X}_p &= \left. \frac{d}{dt} \right|_{t=0} (g(pe^{tX})) \\ &= \left. \frac{d}{dt} \right|_{t=0} (g(p)e^{tX}) \\ &= \left. \frac{d}{dt} \right|_{t=0} (L_{g(p)}e^{tX}) \\ &= (L_g)_*X \end{aligned}$$

□

The algebraic nemesis to the Ehresmann connection is the **connection 1-form**. It is a 1-form $A \in \Omega^1(P, \mathfrak{g})$ on the principal bundle P satisfying,

(1) $R_g^*A = \mathrm{ad}_{g^{-1}} \circ A$ for all $g \in G$;

(2) $A(\tilde{X}) = X$ for every $X \in \mathfrak{g}$.

Breaking this definition down, the connection A being an element in $\Omega^1(P, \mathfrak{g})$ implies at any point $p \in P$ it is the linear map $A_p : T_pP \rightarrow \mathfrak{g}$. Recall the adjoint representation $\mathrm{ad}_{g^{-1}}$ is an automorphism on $\mathfrak{g}$, which shows that $\mathrm{ad}_{g^{-1}} \circ A$ is well-defined and an element of $\Omega^1(P, \mathfrak{g})$.

We can show these two contrasting definitions are actually equivalent by the following theorem.

THEOREM 2.5. Let $(P, \pi, M : G)$ be a principal G -bundle.

(i) For any Ehresmann connection H_p on P we have a connection 1-form A defined by

$$A_p(\tilde{X}_p + Y_p) = X$$

for all $p \in P$, $X \in \mathfrak{g}$ and $Y_p \in H_p$.

(ii) For a connection 1-form $A \in \Omega^1(P, \mathfrak{g})$, we can define an Ehresmann connection H on P by

$$H_p = \ker(A_p)$$

Proof. (i) From H_p to A .

We want to verify that the form A satisfies the defining property of a connection one-form on the principal bundle. Specifically, for every $X \in \mathfrak{g}$, the following holds:

$$A(\tilde{X}) = X,$$

where $\tilde{X}$ is the fundamental vector field on the total space corresponding to X .

Our goal is to compute the pull-back R_g^*A of the form A under the right action of $g \in G$ on the bundle.

We have that the push-forward of the fundamental vector field under right multiplication satisfies

$$(R_g)_*\tilde{X} = \tilde{Z},$$

(ii) From A to H_p . Let A satisfy (i)–(ii), and define

$$H_p^A := \ker(A_p) \subset T_pP.$$

We claim $T_pP = V_p \oplus H_p^A$. First, $H_p^A \cap V_p = \{0\}$: if $v \in V_p$ and $A(v) = 0$, then (1) says $A_p|_{V_p} = \sigma_p^{-1}$, hence $v = 0$. Second, $A_p|_{V_p} = \sigma_p^{-1}$ is an isomorphism by (2), so $\dim H_p^A = \dim T_pP - \dim V_p$, proving it is a complementary subspace. Smoothness of H^A follows from smoothness of ω .

G -invariance: for $g \in G$,

$$A_{pg}((R_g)_*v) = \text{ad}_{g^{-1}}(A_p(v)) \quad \text{by (2)}.$$

Thus $A_p(v) = 0 \iff A_{pg}((R_g)_*v) = 0$, i.e. $(R_g)_*H_p^A = H_{pg}^A$. $\square$

2.6. Curvature Symphony

Let $P \rightarrow M$ be a principal bundle with structure group G and Lie algebra $\mathfrak{g}$. Suppose A is a connection one-form on P . To capture how the connection behaves under infinitesimal loops, we begin by projecting vector fields onto the horizontal distribution.

Denote by π_H the projection of the tangent bundle TP onto the horizontal sub-bundle determined by ω . For any vector fields X, Y on P , define a $\mathfrak{g}$ -valued 2-form F on P by

$$F(X, Y) = d\omega(\pi_H X, \pi_H Y).$$

DEFINITION 2.6. Given a connection form A , the associated **curvature two-form**, $F \in \Omega^2(P, \mathfrak{g})$, is defined as

$$F(X, Y) = dA(\pi_H(X), \pi_H(Y)) \quad \forall X, Y \in T_pP, p \in P,$$

where $\pi_H : TP \rightarrow H$ denotes the projection onto the horizontal subspace. We sometimes write F^A to make explicit that the curvature depends on the chosen connection A .

This form possesses some important structural properties. For instance, it behaves naturally under the action of the group G and vanishes when contracted with a fundamental vector field.

PROPOSITION 2.7. *The curvature F satisfies:*

- (a) *For each $g \in G$, $(r_g)^*F = \text{Ad}_{g^{-1}}F$.*
- (b) *For every $X \in \mathfrak{g}$, $\tilde{X} \lrcorner F = 0$, where $\tilde{X}$ is the fundamental vector field associated with X .*

Proof. (a) Let $X, Y \in T_pP$. By definition,

$$((R_g)^*F)_p(X, Y) = F_{pg}((R_g)_*X, (R_g)_*Y).$$

Since R_g preserves the horizontal distribution, we have $\pi_H((R_g)_*X) = (R_g)_*\pi_H(X)$. Hence

$$((R_g)^*F)_p(X, Y) = dA_{pg}((R_g)_*\pi_H(X), (R_g)_*\pi_H(Y)).$$

Using the equivariance of A , $(R_g)^*A = \text{Ad}_{g^{-1}}A$, and linearity of d , we find

$$((R_g)^*F)_p(X, Y) = \text{Ad}_{g^{-1}}(dA_p(\pi_H(X), \pi_H(Y))) = (\text{Ad}_{g^{-1}}F)_p(X, Y).$$

(b) For $X \in \mathfrak{g}$, its fundamental vector field $\tilde{X}$ is vertical, hence $\pi_H(\tilde{X}) = 0$. For any $Y \in T_pP$,

$$F(\tilde{X}, Y) = dA(\pi_H(\tilde{X}), \pi_H(Y)) = dA(0, \pi_H(Y)) = 0.$$

Thus $\iota_{\tilde{X}}F = 0$. □

Before rewriting F more compactly, we need a way to combine $\mathfrak{g}$ -valued forms using the Lie algebra bracket. Let $\eta \in \Omega^k(P, \mathfrak{g})$ and $\phi \in \Omega^\ell(P, \mathfrak{g})$. We define their bracket $[\eta, \phi] \in \Omega^{k+\ell}(P, \mathfrak{g})$ by

$$[\eta, \phi](X_1, \dots, X_{k+\ell}) = \sum_{\sigma} \text{sgn}(\sigma) [\eta(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \phi(X_{\sigma(k+1)}, \dots, X_{\sigma(k+\ell)})],$$

where the sum runs over all (k, ℓ) -shuffles. In words, this means we antisymmetrize over all arguments as in the usual wedge product, but instead of multiplying values we apply the Lie bracket in $\mathfrak{g}$. Thus $[\eta, \phi]$ behaves like the wedge product, with the algebra replaced by the bracket structure.

In the most common situation, where both forms are Lie algebra-valued one-forms, this simplifies considerably. For $\eta, \phi \in \Omega^1(P, \mathfrak{g})$ we obtain

$$[\eta, \phi](X, Y) = [\eta(X), \phi(Y)] - [\eta(Y), \phi(X)].$$

In particular, setting $\phi = \eta$ yields

$$[\eta, \eta](X, Y) = 2[\eta(X), \eta(Y)].$$

With this preparation in place, we can state a fundamental results in the theory of connections. This identity, called the *structure equation*, provides the compact form of curvature. It reveals that curvature arises from the exterior derivative of the connection one-form together with the Lie bracket interaction of ω with itself.

THEOREM 2.8 Structure Equation. *The curvature of a connection form A is given by*

$$F = dA + \frac{1}{2}[A, A].$$

Proof. Let X, Y be tangent vectors.

Case 1: Both horizontal. Then $F(X, Y) = dA(X, Y)$ by definition. Since $A(X) = A(Y) = 0$, the bracket term vanishes, and $dA + \frac{1}{2}[A, A]$ reduces to dA . Equality follows.

Case 2: Both vertical. Then $\pi_H(X) = \pi_H(Y) = 0$, so $F(X, Y) = 0$.

On the other hand,

$$dA(X, Y) = X(A(Y)) - Y(A(X)) - A([X, Y]),$$

but since $A(X), A(Y)$ are constants in g , this reduces to $-[A(X), A(Y)]$. Adding

$$\frac{1}{2}[A, A](X, Y) = [A(X), A(Y)]$$

, we obtain zero. Hence both sides agree.

Case 3: One vertical, one horizontal. Say $X = \tilde{V}$ is vertical and Y horizontal.

Then $F(X, Y) = 0$. On the other side, $dA(X, Y) = X(A(Y)) - Y(A(X)) - A([X, Y])$. Now $A(Y) = 0$ and $A(X) = V$ is constant. By Lemma 2.7, $[X, Y]$ is horizontal, so $A([X, Y]) = 0$. Hence $dA(X, Y) = 0$. Also $[A, A](X, Y) = [V, 0] = 0$. Thus equality holds.

Therefore in all cases $F = dA + \frac{1}{2}[A, A]$. $\square$

An additional and profound relation involving curvature is provided by the Bianchi identity. This result demonstrates that the curvature two-form is not arbitrary but satisfies a natural differential constraint.

THEOREM 2.9. *For the curvature two-form F of a connection A ,*

$$dF = 0 \quad \text{on } H \times H \times H.$$

Proof. We use the graded Leibniz rule. Starting with $F = dA + \frac{1}{2}[A, A]$, we compute

$$dF = d^2A + \frac{1}{2}d[A, A].$$

The first term vanishes since $d^2 = 0$. Using the Leibniz rule for the graded bracket gives $d[A, A] = 2[dA, A]$. Substituting, we get

$$dF = [dA, A].$$

But on horizontals, $dA = F - \frac{1}{2}[A, A]$, and since A vanishes on horizontals, this reduces to $[F, A] = 0$. Hence $dF = 0$ on horizontal triples. $\square$

The curvature 2-form records the failure of the horizontal distribution to be integrable. Suppose X and Y are horizontal vector fields on P . While each of X and Y is tangent to the horizontal distribution, their Lie bracket $[X, Y]$ need not remain horizontal. The vertical component of $[X, Y]$ is precisely captured by the curvature:

$$F(X, Y) = -A([X, Y]).$$

In this sense, curvature measures the “twisting” of horizontal subspaces as one moves in different directions. If F vanishes identically, the horizontal distribution is integrable, meaning the bundle admits local sections with flat horizontal structure. Otherwise, the curvature encodes the precise obstruction to such integrability.

The curvature 2-form provides a natural, coordinate-free object that links the connection on a principal bundle with its global geometry. It transforms covariantly, vanishes on vertical directions, and satisfies a universal identity (the Bianchi identity).

2.7. Parallel Transport

Parallel transport provides a way to lift curves from the base manifold M to the total space P along horizontal directions. Specifically, given a smooth curve $\gamma : [0, 1] \rightarrow M$ and a point $p_0 \in P$ such that $\pi(p_0) = \gamma(0)$, there exists a unique curve $\tilde{\gamma} : [0, 1] \rightarrow P$ satisfying:

- (1) $\tilde{\gamma}(0) = p_0$,
- (2) $\pi \circ \tilde{\gamma} = \gamma$,
- (3) $\dot{\tilde{\gamma}}(t) \in H_{\tilde{\gamma}(t)}P$ for all $t \in [0, 1]$.

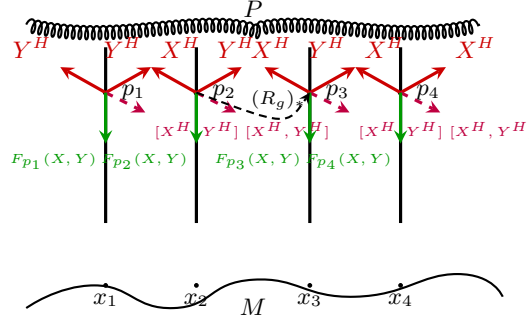


FIGURE 4. Curvature as seen across several fibres: at each $p_i \in \pi^{-1}(x_i)$ horizontal lifts X^H, Y^H have Lie bracket whose vertical component is precisely $F_{p_i}(X, Y)$.

This curve $\tilde{\gamma}$ is called the *horizontal lift* of γ through p_0 . The endpoint $\tilde{\gamma}(1)$ depends smoothly on p_0 and γ and defines the *parallel transport map* along γ in P .

Parallel transport along infinitesimal curves allows one to compare points in different fibres in a manner compatible with the bundle structure. Because horizontal subspaces are preserved under the right G -action, parallel transport is equivariant:

$$\tilde{\gamma}_{pg} = \tilde{\gamma}_p \cdot g,$$

where $\tilde{\gamma}_p$ denotes the horizontal lift starting at p .

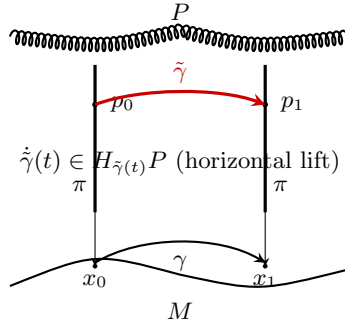


FIGURE 5. Parallel transport along a curve γ in the base: the unique horizontal lift $\tilde{\gamma}$ starting at p_0 projects to γ and defines the transported point p_1 .

2.8. Covariant Derivative

Associated with a connection on P is a natural notion of differentiation along vector fields on M . Let $E \rightarrow M$ be a smooth vector bundle associated to P via a representation $\rho : G \rightarrow \text{GL}(V)$. A smooth section $s \in \Gamma(E)$ can be lifted to an equivariant function $\tilde{s} : P \rightarrow V$. The *covariant derivative* of s in the direction of a vector field $X \in \mathfrak{X}(M)$ is defined by

$$\nabla_X s(x) := \left. \frac{d}{dt} \right|_{t=0} \tilde{s}(\tilde{\gamma}(t)),$$

where $\tilde{\gamma}$ is the horizontal lift of any curve γ in M satisfying $\gamma(0) = x$ and $\dot{\gamma}(0) = X_x$. Equivalently, $\nabla_X s$ measures the rate of change of s along X while remaining horizontal with respect to the connection.

The covariant derivative satisfies the usual properties:

- (1) Linearity in X and s ,
- (2) Leibniz rule: $\nabla_X(fs) = (Xf)s + f\nabla_X s$ for $f \in C^\infty(M)$,
- (3) Compatibility with the bundle structure: it respects the equivariance induced by the principal bundle action.

In this way, the connection on P induces a well-defined method for differentiating sections of associated bundles, providing a geometric framework for transporting data along curves in the base manifold while remaining compatible with the bundle structure.

3. The Hopf Fibration

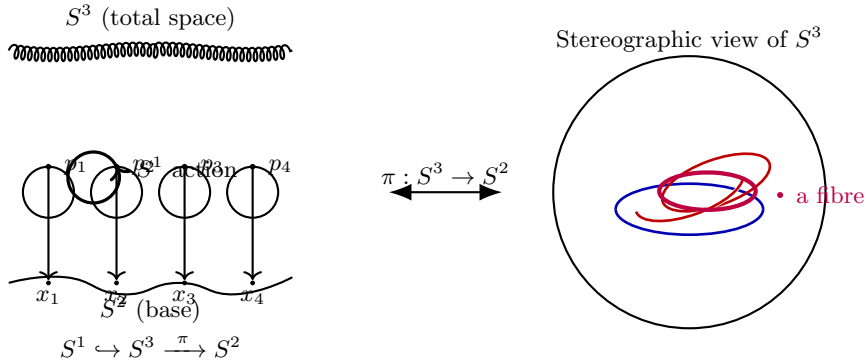


FIGURE 6. Left: Hopf fibration as a principal S^1 -bundle with vertical fibres and S^1 -action. Right: stereographic-style view of S^3 showing several Hopf fibres as linked circles.

We dedicate this section to a major principal G-bundle, that was in fact accidentally discovered by Heinz Hopf (1894-1971) when he was studying continuous maps between spheres of different dimensions in 1931, and their homotopy groups. Coincidentally, that same year, Dirac discovered his magnetic monopole whose fibre bundle turns out to be the Hopf fibration itself, but these ideas were only linked in the 1970s. This structure has found other significant applications in physics including understanding rigid body mechanics, and visualizing quantum mechanical states through the Bloch sphere. However, our interests lie in its deep geometric structure revealed by deconstructing a 3-sphere into a collection of circles arranged on a 2-sphere. The intention is to conceptualise the Hopf fibration in relation to principal bundle terminology.

Our base space M is going to be the complex projective line $\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{0\}) / \sim$ where $\sim$ is the equivalence relation defined as

$$(w_1, z_1) \sim (w_2, z_2) \Leftrightarrow (w_1, z_1) = (tw_2, tz_2)$$

for some $t \in \mathbb{C}$.

The set of equivalence classes will be denoted as $[w, z]$. Altogether we have $\mathbb{CP}^1 = \{[w : 1] : z \in \mathbb{C}\} \cup [1 : 0]$.

Alternatively, the complex projective line $\mathbb{CP}^1$ can be thought of as the *one-point compactification*. Imagine the complex plane as an infinite sheet that extends endlessly in all directions, then wrap this plane onto a sphere by bringing this point at infinity and connecting it smoothly to the rest of the plane, so the entire extended plane fits perfectly onto the surface of a 2-sphere. This "wrapping" corresponds to stereographic projection: every point on the sphere except the north pole is projected onto the plane, and the north pole itself represents the point at infinity. Conversely, we can go from the plane to the sphere by sending each point

to a unique point on the sphere, and the added point at infinity to the north pole. Because $\mathbb{CP}^1$ is precisely the complex plane plus this point at infinity, topologically it is the same as a sphere. The natural coordinate charts covering $\mathbb{CP}^1$ correspond to two hemispheres of this sphere joined along their equator via the inversion map $z \mapsto 1/z$ which matches with how the sphere's northern and southern hemispheres are glued in stereographic coordinates. Thus, by this geometric intuition, the complex projective line is diffeomorphic to the 2-sphere. So using this brief topological argument, we can characterise our base space as the 2-sphere, S^2 .

We take the 1-sphere, (or more commonly known as the circle), to be the following

$$S^1 = \{e^{i\theta} \in \mathbb{C} : \theta \in \mathbb{R}\}.$$

It is an abelian group closed under multiplication, and the verification of the other three group axioms is a simple exercise. Furthermore, S^1 qualifies as a Lie group which we demonstrate directly from definitions. The two maps

$$\begin{aligned} \mathbb{C} \times \mathbb{C} &\rightarrow \mathbb{C} \\ (e^{i\theta}, e^{i\xi}) &\mapsto e^{i(\theta+\xi)} \\ \mathbb{C} \setminus \{0\} &\rightarrow \mathbb{C} \setminus \{0\} \\ e^{i\theta} &\mapsto e^{-i\theta} \end{aligned}$$

representing multiplication and the inverse respectively, are smooth mappings with restrictions to S^1 , implying the two group operations are smooth. Adhering to customs, we denote S^1 by its official Lie group name, the unitary group of degree 1, or just $U(1)$. Sticking to terminology, this is the general fibre of the Hopf fibration.

The principal $U(1)$ -bundle is the *hypersphere*, or the 3-sphere, S^3 , given by

$$S^3 := \{(w, z) \in \mathbb{C}^2 : |w|^2 + |z|^2 = 1\}.$$

THEOREM 3.1. *The Hopf fibration*

$$\pi : S^3 \rightarrow S^2$$

is a principal $U(1)$ -bundle.

Proof.

Step 1: The Hopf map. Define

$$\pi : S^3 \longrightarrow \mathbb{CP}^1, \quad \pi(w, z) = [w : z].$$

Since $\mathbb{CP}^1 \cong S^2$, this realizes the Hopf fibration $S^3 \rightarrow S^2$.

Step 2: The $U(1)$ -action. Let $U(1)$ act on S^3 by

$$\lambda \cdot (w, z) = (\lambda w, \lambda z) \quad \lambda \in U(1)$$

This action is smooth. If $\lambda \cdot (w, z) = (w, z)$ then $\lambda w = w$ and $\lambda z = z$. Since $(w, z) \neq (0, 0)$, we conclude $\lambda = 1$. Thus the action is free. The fibres of π are exactly the $U(1)$ -orbits.

Step 3: Local trivializations. Consider the standard affine cover of $\mathbb{CP}^1$:

$$U_0 = \{[w : z] \mid w \neq 0\}, \quad U_1 = \{[w : z] \mid z \neq 0\}.$$

On U_0 , set $v = z/w$. Define a smooth section

$$s_0(v) = \frac{(1, v)}{\sqrt{1 + |v|^2}} \in S^3.$$

For any $(w, z) \in \pi^{-1}(U_0)$ we may write

$$v = \frac{z}{w}, \quad \lambda = \frac{w}{|w|} \in U(1),$$

and one checks

$$(w, z) = \lambda \cdot s_0(v).$$

Thus we obtain the trivialization

$$\Phi_0 : \pi^{-1}(U_0) \longrightarrow U_0 \times U(1), \quad \Phi_0(w, z) = (v, \lambda).$$

Similarly, on U_1 with coordinate $v' = w/z$, define

$$s_1(v') = \frac{(v', 1)}{\sqrt{1 + |v'|^2}},$$

and obtain the trivialization

$$\Phi_1 : \pi^{-1}(U_1) \longrightarrow U_1 \times U(1).$$

Step 4: Transition functions. On the overlap $U_0 \cap U_1$, with $v = z/w$ and $v' = 1/v$, the transition between trivializations is

$$(v, \lambda) \mapsto \left(v', \lambda \cdot \frac{v}{|v|}\right),$$

which is smooth and takes values in $U(1)$.

Step 5: Conclusion. We have exhibited:

- a smooth free $U(1)$ -action on S^3 with orbits the fibres of π ;
- local trivializations Φ_0, Φ_1 which are $U(1)$ -equivariant.

Therefore, $\pi : S^3 \rightarrow \mathbb{CP}^1 \cong S^2$ is a principal $U(1)$ -bundle. □

3.1. The connection 1-form on the Hopf Fibration

To construct a connection, we begin with the tangent spaces of S^3 . At a point $(w, z) \in S^3$, the tangent space is

$$T_{(w,z)}S^3 = \{(X_1, X_2) \in \mathbb{C}^2 : \bar{w}X_1 + \bar{z}X_2 = 0\}.$$

It is convenient to introduce 1-forms $\alpha_j, \bar{\alpha}_j \in \Omega^1(S^3, \mathbb{C})$ defined by

$$\alpha_j(X_1, X_2) = X_j, \quad \bar{\alpha}_j(X_1, X_2) = \bar{X}_j.$$

In terms of these, one can write down the canonical connection 1-form on S^3 . Explicitly, it is given by

$$A_{(w,z)}(X_1, X_2) = \frac{1}{2} (\bar{w}X_1 - w\bar{X}_1 + \bar{z}X_2 - z\bar{X}_2).$$

This is a real-valued 1-form on S^3 with values in the Lie algebra $\mathfrak{u}(1) \cong i\mathbb{R}$.

To check that A really is a connection 1-form for the Hopf bundle, two properties must hold:

- (1) *Equivariance under the $U(1)$ action.* Since $U(1)$ is abelian, we require simply

$$(R_g^*A)(X) = A(X), \quad \text{for all } g \in U(1), X \in TS^3,$$

where R_g denotes the right action of g . A straightforward computation using $|w|^2 + |z|^2 = 1$ confirms this invariance.

- (2) *Reproduction of vertical directions.* For a Lie algebra element $Y \in \mathfrak{u}(1)$, let $\tilde{Y}$ denote the corresponding fundamental vertical vector field. Then A must satisfy

$$A(\tilde{Y}) = Y.$$

Writing $Y = iy$ with $y \in \mathbb{R}$, the associated fundamental vector field is

$$\tilde{Y}_{(w,z)} = \frac{d}{dt} \Big|_{t=0} (we^{ity}, ze^{ity}) = (iyw, iyz).$$

Evaluating the connection form gives

$$A_{(w,z)}(\tilde{Y}) = \frac{1}{2}(\bar{w}(iyw) + w(-iy\bar{w}) + \bar{z}(iyz) + z(-iy\bar{z})) = iy(|w|^2 + |z|^2) = Y,$$

since $(w, z) \in S^3$ satisfies $|w|^2 + |z|^2 = 1$.

Thus A defines a genuine $U(1)$ -connection 1-form on the Hopf fibration. Geometrically, it selects horizontal directions in the tangent bundle of S^3 complementary to the vertical directions generated by the $U(1)$ action.

3.2. Curvature of the connection on the Hopf fibration

We now examine the curvature of its natural $U(1)$ -invariant connection. Since the structure group is abelian, the general structure equation simplifies: for a connection 1-form A we have

$$F^A = dA,$$

because the quadratic term $\frac{1}{2}[A \wedge A]$ vanishes identically. Thus, the task reduces to computing the exterior derivative of A .

One convenient way to describe A is in terms of the standard complex coordinates on $\mathbb{C}^2$, where S^3 is given by $\{|w|^2 + |z|^2 = 1\}$. Writing

$$A = \frac{i}{2}(\bar{w}dw - w d\bar{w} + \bar{z}dz - z d\bar{z}),$$

we obtain, after differentiation,

$$F^A = -(\alpha_1 \wedge \bar{\alpha}_1 + \alpha_2 \wedge \bar{\alpha}_2),$$

where the 1-forms α_i and $\bar{\alpha}_i$ denote the standard complex components of dz_i and $d\bar{z}_i$ restricted to S^3 . This explicit expression shows that F^A is non-trivial but globally well defined and invariant under the $U(1)$ action.

A fundamental property of this curvature form is that it descends to the base S^2 . Indeed, by pulling back along a local section $s : S^2 \rightarrow S^3$, one obtains a 2-form F_{S^2} on S^2 satisfying

$$F_{S^2} \Big|_v = s^* F^A.$$

Thus the connection on the Hopf fibration determines a canonical closed 2-form on the sphere.

To see the topological significance of this form, we compute its integral. Since $S^2 \simeq \mathbb{CP}^1$, the local affine coordinate $w \in \mathbb{C}$ provides a chart in which

$$F_{S^2} = \frac{1}{(1 + |w|^2)^2} dw \wedge d\bar{w}.$$

Switching to polar coordinates $w = re^{i\phi}$, a straightforward calculation shows

$$\int_{S^2} F_{S^2} = 2\pi i.$$

Normalising by $\frac{1}{2\pi i}$, this integral equals 1.

It is customary to write the normalised form

$$\omega_{FS} = \frac{1}{2i} F_{S^2},$$

which is the Fubini–Study Kähler form on $\mathbb{CP}^1$. In fact, one can show

$$\omega_{FS} = \frac{1}{4} \omega_{\text{std}},$$

where ω_{std} is the standard area form on the round 2-sphere of radius 1. In this way, the curvature of the connection on the Hopf fibration is geometrically identified with the Fubini-Study metric on $\mathbb{CP}^1$.

When dealing with Lie algebra-valued forms, the wedge product interacts with the Lie bracket in a natural way. If ω is a $\mathfrak{g}$ -valued 1-form, then for vector fields X, Y we define

$$[\omega \wedge \omega](X, Y) = [\omega(X), \omega(Y)]_{\mathfrak{g}}.$$

Here the wedge product antisymmetrises the arguments (X, Y) , while the multiplication on the Lie algebra side is given by the Lie bracket. This construction ensures that expressions such as the structure equation

$$F = d\omega + \frac{1}{2}[\omega \wedge \omega]$$

make sense. In the abelian case (e.g. $\mathfrak{u}(1)$), the Lie bracket vanishes, and the formula reduces to $F = d\omega$.

4. G -Invariant Connection 1-forms and Curvature 2-forms on $\text{Fr}(P)$

Now we investigate the conditions under which a connection on the frame bundle of a principal bundle is invariant under the action of a Lie group. Motivated by the Hopf fibration, $S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$, we explore how a free group G on a principal bundle P lifts to an action on the frame bundle, and how one may construct a G -invariant connection on $\text{Fr}(M)$.

4.1. $U(1)$ -Invariant Connections and Curvature on $\text{Fr}(S^3)$

Consider the Hopf fibration

$$S^1 \cong U(1) \hookrightarrow S^3 \xrightarrow{\pi} S^2,$$

where the structure group $U(1)$ acts on $S^3 \subset \mathbb{C}^2$ by complex multiplication:

$$R_z : S^3 \rightarrow S^3, \quad R_z(p) = p \cdot z, \quad z \in U(1).$$

By definition, this is a right principal bundle action.

Let $\text{Fr}(S^3)$ denote the frame bundle of S^3 , whose points are pairs $(p, (e_1, \dots, e_3))$ where $p \in S^3$ and $(e_1, \dots, e_3)$ is an ordered basis of $T_p S^3$. Our goal is to describe how the $U(1)$ -action on S^3 naturally induces an action on $\text{Fr}(S^3)$.

DEFINITION 4.1. For each $\lambda \in U(1)$, define a map

$$\tilde{R}_\lambda : \text{Fr}(S^3) \rightarrow \text{Fr}(S^3), \quad \tilde{R}_\lambda(p, (e_1, \dots, e_3)) := (p \cdot \lambda, ((dR_\lambda)_p e_1, \dots, (dR_\lambda)_p e_3)),$$

where $(dR_\lambda)_p : T_p S^3 \rightarrow T_{p \cdot \lambda} S^3$ is the differential of R_λ at p .

Since $(dR_\lambda)_p$ is a linear isomorphism, it sends any basis of $T_p S^3$ to a basis of $T_{p \cdot \lambda} S^3$, so $\tilde{R}_\lambda$ is well-defined. This construction essentially “pushes forward” the frame along the bundle action.

PROPOSITION 4.2. The map $\tilde{R}_\lambda$ defines a right action of $U(1)$ on $\text{Fr}(S^3)$. Moreover, this action commutes with the standard right action of $GL(3, \mathbb{R})$ on frames, so each $\tilde{R}_\lambda$ is a principal-bundle automorphism of $\text{Fr}(S^3)$ covering $R_\lambda : S^3 \rightarrow S^3$.

Proof. To verify the group law, take $\lambda_1, \lambda_2 \in U(1)$. Then

$$\tilde{R}_{\lambda_1}(\tilde{R}_{\lambda_2}(p, (e_i))) = \tilde{R}_{\lambda_1 \cdot \lambda_2}(p, (e_i)),$$

because $R_{\lambda_1} \circ R_{\lambda_2} = R_{\lambda_1 \cdot \lambda_2}$ and $d(R_{\lambda_1} \circ R_{\lambda_2}) = (dR_{\lambda_1}) \circ (dR_{\lambda_2})$. This confirms that $\tilde{R}$ satisfies the right action property.

Next, recall that $\text{Fr}(S^3)$ carries its own right $GL(3, \mathbb{R})$ -action given by post-composition on frames:

$$(p, (e_i)) \cdot B = (p, (\sum_j e_j B_{ji})_i), \quad B \in GL(3, \mathbb{R}).$$

Since $(dR_\lambda)_p$ is linear, it commutes with post-composition by B :

$$\tilde{R}_\lambda((p, (e_i)) \cdot B) = \tilde{R}_\lambda(p, (e_i)) \cdot B.$$

Thus the induced $U(1)$ -action is compatible with the frame-bundle structure and preserves the principal $GL(3, \mathbb{R})$ -bundle structure. $\square$

REMARK 1. The induced action $\tilde{R}$ preserves any $U(1)$ -invariant geometric structure on S^3 , such as a metric, a horizontal distribution, or a connection. In particular, if we later define a $U(1)$ -invariant connection 1-form on the Hopf bundle, its horizontal distribution will be invariant under the push-forward action on frames.

4.2. Invariant connections on the frame bundle: forms vs. Ehresmann distributions

Let $\pi : \text{Fr}(S^3) \rightarrow S^3$ be the frame bundle of S^3 , a principal $GL(3, \mathbb{R})$ -bundle. Recall from the previous section that the Hopf $U(1)$ -action on S^3 induces a right action

$$\tilde{R}_\lambda : \text{Fr}(S^3) \rightarrow \text{Fr}(S^3), \quad \lambda \in U(1),$$

given by push-forward of frames. In the sequel we write $P = \text{Fr}(S^3)$ and denote the Lie algebra of $GL(3, \mathbb{R})$ by $\mathfrak{gl}(3, \mathbb{R})$.

The goal is to define one, what it means for a connection 1-form on P to be $U(1)$ -invariant; two, what it means for an Ehresmann connection to be $U(1)$ -invariant; and three, prove these notions are equivalent by establishing a one-to-one correspondence between $U(1)$ -invariant connection forms and $U(1)$ -invariant Ehresmann connections.

DEFINITION 4.3. A connection 1-form ω on P is $U(1)$ -**invariant** (with respect to the induced action $\tilde{R}$) if for every $\lambda \in U(1)$

$$\tilde{R}_\lambda^* \omega = \omega.$$

DEFINITION 4.4. An Ehresmann connection H is $U(1)$ -**invariant** if for every $\lambda \in U(1)$ and every $p \in P$

$$(d\tilde{R}_\lambda)_p(H_p) = H_{p \cdot \lambda}.$$

That is, the induced action $\tilde{R}_\lambda$ preserves the horizontal distribution.

These two definitions capture the same underlying geometric idea: invariance means that parallel transport and the splitting of tangent directions into “horizontal” and “vertical” are unaffected by the $U(1)$ symmetry. The connection does not distinguish between points related by the symmetry; in other words, the geometry it encodes is compatible with the group action.

THEOREM 4.5. There is a one-to-one correspondence between $U(1)$ -invariant connection 1-forms ω on $P = \text{Fr}(S^3)$ and $U(1)$ -invariant Ehresmann connections $H \subset TP$. Concretely:

- i If ω is a connection form with $\tilde{R}_\lambda^* \omega = \omega$ for all $\lambda \in U(1)$, then $H = \ker \omega$ is an $U(1)$ -invariant Ehresmann connection.
- ii Conversely, if H is an $U(1)$ -invariant Ehresmann connection, there exists a unique connection 1-form ω with $H = \ker \omega$ and $\tilde{R}_\lambda^* \omega = \omega$ for all $\lambda \in U(1)$.

Proof. We split the proof into two directions.

(i) *From invariant connection form to invariant horizontal distribution.* Let ω be a connection 1-form on P satisfying $\tilde{R}_\lambda^* \omega = \omega$ for every $\lambda \in U(1)$. Define $H_p := \ker(\omega_p) \subset T_p P$. Standard properties of connection forms give $T_p P = H_p \oplus V_p$ and H is $GL(3, \mathbb{R})$ -equivariant because ω satisfies the $GL(3, \mathbb{R})$ -equivariance property $R_g^* \omega = \text{Ad}_{g^{-1}} \omega$. It remains to show H is preserved by the induced $U(1)$ -action. For any $\lambda \in U(1)$ and $X \in H_p$ we compute

$$\begin{aligned} \omega_{p \cdot \lambda}((d\tilde{R}_\lambda)_p X) &= (\tilde{R}_\lambda^* \omega)_p(X) \\ &= \omega_p(X) \\ &= 0, \end{aligned}$$

so $(d\tilde{R}_\lambda)_p X \in \ker \omega_{p \cdot \lambda} = H_{p \cdot \lambda}$. Thus $(d\tilde{R}_\lambda)_p(H_p) \subset H_{p \cdot \lambda}$; applying the same argument to $\tilde{R}_{\lambda^{-1}}$ shows equality. Therefore H is $U(1)$ -invariant, as required.

(ii) *From invariant horizontal distribution to invariant connection form.* Suppose $H \subset TP$ is an Ehresmann connection which is $U(1)$ -invariant. We first recall the standard construction of the connection form from H : each tangent vector $X \in T_p P$ can be uniquely decomposed as

$$X = X^H + X^V, \quad X^H \in H_p, \quad X^V \in V_p.$$

There is a canonical linear isomorphism

$$\mathfrak{gl}(3, \mathbb{R}) \xrightarrow{\cong} V_p, \quad \tilde{X} \mapsto \tilde{X}_P(p),$$

where $\tilde{X}_P$ denotes the fundamental vertical vector field associated to $\tilde{X}$. Using this isomorphism we define $\omega_p : T_p P \rightarrow \mathfrak{gl}(3, \mathbb{R})$ by

$$\omega_p(X) = \tilde{X} \quad \text{iff} \quad X^V = \tilde{X}_P(p).$$

Equivalently, ω_p is the projection onto the vertical part followed by identification with the Lie algebra via the fundamental field map. Standard arguments show ω is a smooth $\mathfrak{gl}(3, \mathbb{R})$ -valued 1-form and satisfies the reproduction property $\omega(\tilde{X}_P) = \tilde{X}$ and the $GL(3, \mathbb{R})$ -equivariance $R_g^* \omega = \text{Ad}_{g^{-1}} \omega$, so ω is a connection form with $H = \ker \omega$.

It remains to check ω is $U(1)$ -invariant. Fix $\lambda \in U(1)$ and $p \in P$. For any $X \in T_p P$ write $X = X^H + X^V$ as above. Since H is $U(1)$ -invariant we have $(d\tilde{R}_\lambda)_p(X^H) \in H_{p \cdot \lambda}$, while $(d\tilde{R}_\lambda)_p(X^V) \in V_{p \cdot \lambda}$. Moreover, the identification $\mathfrak{gl}(3, \mathbb{R}) \cong V_p$ intertwines the differential of bundle automorphisms with the adjoint action: for the right GL -action one has R_g -equivariance, and for $\tilde{R}_\lambda$ (which is a bundle automorphism covering $R_\lambda : S^3 \rightarrow S^3$ and commuting with the GL -action) the push-forward of a fundamental field satisfies

$$(d\tilde{R}_\lambda)_p(\tilde{X}_P(p)) = \tilde{X}_P(p \cdot \lambda), \quad \forall \tilde{X} \in \mathfrak{gl}(3, \mathbb{R}).$$

Thus, if $\omega_p(X) = \tilde{X}$ (so $X^V = \tilde{X}_P(p)$), then

$$\begin{aligned} (\tilde{R}_\lambda^* \omega)_p(X) &= \omega_{p \cdot \lambda}((d\tilde{R}_\lambda)_p X) \\ &= \omega_{p \cdot \lambda}((d\tilde{R}_\lambda)_p X^H + (d\tilde{R}_\lambda)_p X^V) \\ &= \omega_{p \cdot \lambda}((d\tilde{R}_\lambda)_p X^V). \end{aligned}$$

The last equality holds because $(d\tilde{R}_\lambda)_p X^H$ is horizontal and thus annihilated by $\omega_{p \cdot \lambda}$. Using the intertwining of fundamental fields we get

$$\begin{aligned}\omega_{p \cdot \lambda}((d\tilde{R}_\lambda)_p X^V) &= \omega_{p \cdot \lambda}(\tilde{X}_P(p \cdot \lambda)) \\ &= \tilde{X}.\end{aligned}$$

Hence $(\tilde{R}_\lambda^* \omega)_p(X) = \tilde{X} = \omega_p(X)$ for all X , so $\tilde{R}_\lambda^* \omega = \omega$. Because $\lambda \in U(1)$ was arbitrary, ω is $U(1)$ -invariant. Uniqueness of ω with kernel H is immediate from the definition (the vertical projection is uniquely determined by H), completing the proof. $\square$

One question is why might such arguments exist in general? Well the equivalence just proved is not special to the Hopf fibration or, to $GL(3, \mathbb{R})$: it is the standard and general correspondence between connection 1-forms and Ehresmann connections on any principal K -bundle. The extra symmetry (here the $U(1)$ -action) acts by bundle automorphisms, and bundle automorphisms naturally pull back forms and push forward distributions. Therefore any invariance condition for forms translates exactly into the corresponding invariance condition for distributions. In abstract terms:

- The assignment $\omega \mapsto \ker \omega$ and its inverse are canonical and functorial constructions. They are compatible with pull-back by bundle automorphisms.
- If a Lie group G acts on P by principal-bundle automorphisms (covering some base action), then G -invariance of ω is equivalent to G -invariance of the associated horizontal distribution, because pull-back of ω along the automorphism corresponds to push-forward of the kernel subspaces.

Consequently, the same proof works for any principal bundle $P \rightarrow M$ and any group G acting by bundle automorphisms: G -invariant connection forms $\longleftrightarrow$ G -invariant Ehresmann connections. This functorial viewpoint is why textbooks (e.g. Kobayashi–Nomizu) treat the two perspectives as interchangeable and why a correspondence theorem such as the one we referred to in Hamilton appears naturally in many texts.

REMARK 2. In practice, when working with the Hopf fibration one often uses the explicit connection 1-form $\alpha = \Im(\bar{z}_1 dz_1 + \bar{z}_2 dz_2)$ on S^3 and the induced connection on associated bundles. The invariance of α under the $U(1)$ -action is immediate from its definition, and the argument above shows that the horizontal distribution it defines is the unique $U(1)$ -invariant Ehresmann connection with that kernel.

4.3. Invariant curvature forms in the Hopf Fibration

Once a $U(1)$ -invariant connection has been fixed on the frame bundle $\text{Fr}(S^3)$, the next natural object to study is its curvature. Again, geometrically, curvature measures the failure of horizontal distributions to be integrable. From the algebraic viewpoint, it is defined via the Maurer–Cartan structure equation applied to the connection 1-form.

DEFINITION 4.6. Given a connection 1-form $\omega \in \Omega^1(\text{Fr}(S^3); \mathfrak{gl}(3, \mathbb{R}))$, the associated curvature 2-form is $F \in \Omega^2(\text{Fr}(S^3); \mathfrak{gl}(3, \mathbb{R}))$

$$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega].$$

DEFINITION 4.7. We say that a curvature 2-form $F \in \Omega^2(S^3, \mathfrak{gl}(3, \mathbb{R}))$ is $U(1)$ -**invariant** if for every $\lambda \in U(1)$,

$$(R_\lambda)^* F = F,$$

where R_λ is the right $U(1)$ -action on $\text{Fr}(S^3)$.

This definition is natural because the curvature is derived functorially from the connection. In fact, if ω is a $U(1)$ -invariant connection 1-form, then the associated curvature 2-form is automatically $U(1)$ -invariant. Indeed, invariance of ω under the group action implies invariance of its exterior derivative $d\omega$, and the wedge bracket $[\omega \wedge \omega]$ is bilinear and respects the pull-back. Thus:

$$(R_\lambda)^* \Omega = d((R_\lambda)^* \omega) + \frac{1}{2} [(R_\lambda)^* \omega \wedge (R_\lambda)^* \omega] = d\omega + \frac{1}{2} [\omega \wedge \omega] = \Omega$$

REMARK 3. From the point of view of parallel transport, invariance of curvature ensures that the infinitesimal holonomy around loops is independent of where the loop is placed along a $U(1)$ -orbit. In the case of the Hopf fibration, this reflects the fact that the fibration is homogeneous: moving along the fibre does not alter the local geometry encoded by the curvature.

Example (Hopf curvature). On the standard Hopf bundle $S^1 \hookrightarrow S^3 \rightarrow S^2$, the natural connection 1-form

$$\alpha = \Im(\bar{z}_1 dz_1 + \bar{z}_2 dz_2)$$

is $U(1)$ -invariant. Its curvature is given by From the perspective of parallel transport, invariance of the curvature means that transporting vectors around small loops produces the same infinitesimal effect, regardless of where the loop is located along a $U(1)$ -orbit. In the case of the Hopf fibration, this expresses the fact that the bundle is homogeneous: moving along the fibre does not change the local geometry encoded in the curvature.] which descends to a multiple of the standard area form on S^2 . By the argument above, F is also $U(1)$ -invariant. This computation provides the simplest concrete illustration of an invariant curvature form in our setting.

4.4. Generalisation to a principal K -bundle with a G -symmetry

We now generalise the Hopf analysis to the following setting, which will be the framework for the remainder of the essay. Let

$$\pi : P \longrightarrow M$$

be a smooth principal K -bundle (K a Lie group) over a manifold M . Assume a Lie group G acts smoothly on P by bundle automorphisms covering an action on M ; write this action either as a right action

$$R_g : P \rightarrow P, \quad p \mapsto p \cdot_G g,$$

or as a left action

$$L_g : P \rightarrow P, \quad p \mapsto g \cdot_G p,$$

depending on one's convention. In particular the action satisfies the bundle-automorphism property

$$\pi(R_g(p)) = g \cdot_M \pi(p) \quad (\text{or } \pi(L_g(p)) = g \cdot_M \pi(p))$$

and it is compatible with the principal right K -action on P in the sense explained below. The aim of this section is to transfer the notions and results we established for the Hopf fibration to this general context: induced actions on frame bundles, definitions of invariance for connection forms, Ehresmann distributions, and curvature, the equivalence between these invariant notions, and the behaviour of parallel transport and covariant derivatives under G .

REMARK 4. There are two separate group actions to keep track of:

- the structure group K of the principal bundle (a right action on P), and
- the symmetry group G (an action by bundle automorphisms which covers an action on M).

The most convenient (and common) situation is when G acts by *principal bundle automorphisms*, i.e. for each $g \in G$ we have

$$R_g(p \cdot k) = R_g(p) \cdot k \quad (\text{or } L_g(p \cdot k) = L_g(p) \cdot k),$$

so that the G -action commutes with the right K -action. In more general situations the G -action may cover an automorphism of K ; then one records a homomorphism $\psi_g : K \rightarrow K$ with

$$\Phi_g(p \cdot k) = \Phi_g(p) \cdot \psi_g(k),$$

and the equivariance conditions in what follows must be modified accordingly. We treat the principal-automorphism case first, and then comment on the more general equivariant case.

Let $\text{Fr}(P)$ denote the linear frame bundle of the manifold P (so $\text{Fr}(P)$ is a principal $GL(\dim P, \mathbb{R})$ -bundle whose fibre over $p \in P$ is the set of ordered bases of $T_p P$). Any diffeomorphism $\Phi : P \rightarrow P$ induces a map on frames by push-forward of bases. Concretely, if Φ is one of the G -action maps ($\Phi = R_g$ or $\Phi = L_g$), define

$$\tilde{\Phi}_g : \text{Fr}(P) \rightarrow \text{Fr}(P), \quad \tilde{\Phi}_g(p, (e_1, \dots, e_n)) = (\Phi_g(p), ((d\Phi_g)_p e_1, \dots, (d\Phi_g)_p e_n)).$$

Because $(d\Phi_g)_p$ is a linear isomorphism $T_p P \rightarrow T_{\Phi_g(p)} P$, $\tilde{\Phi}_g$ is well defined. Moreover the family $\{\tilde{\Phi}_g\}_{g \in G}$ satisfies the same left/right group law as the original action (so a right action on P induces a right action on $\text{Fr}(P)$, and similarly for left actions). Thus the induced action on $\text{Fr}(P)$ inherits the left/right convention exactly.

If G acts by principal-bundle automorphisms (i.e. commutes with the right K -action), then each $\tilde{\Phi}_g$ commutes with the right GL -action on frames (post-composition of frames). If the G -action instead twists the K -action by automorphisms ψ_g , then $\tilde{\Phi}_g$ is still a bundle automorphism of $\text{Fr}(P)$ but its compatibility with structure group actions must be traced through the ψ_g -twist.

REMARK 5.

In working with a principal K -bundle $\pi : P \rightarrow M$ one must distinguish between two different kinds of group actions:

- The *structure group* K , which always acts on the *right*:

$$R_k : P \longrightarrow P, \quad p \mapsto p \cdot k.$$

This convention is part of the definition of a principal bundle and is never altered.

- An additional *symmetry group* G , which we assume acts on P by bundle automorphisms covering an action on M :

$$\Phi_g : P \longrightarrow P.$$

For the G -action there is a choice: it may be defined either as a *left* action

$$\Phi_g \circ \Phi_h = \Phi_{gh},$$

or as a *right* action

$$\Phi_g \circ \Phi_h = \Phi_{hg}.$$

The choice of convention propagates functorially to all induced constructions, including the action on the frame bundle $\text{Fr}(P)$. Thus:

$$\tilde{\Phi}_g(p, (e_1, \dots, e_n)) = (\Phi_g(p), ((d\Phi_g)_p e_1, \dots, (d\Phi_g)_p e_n))$$

defines either a left or right action on $\text{Fr}(P)$ depending only on the convention fixed for the action on P .

The "abelianness" or "non-abelianness" of G plays no role here: if G is abelian, then left and right actions happen to coincide, but in the non-abelian case one simply has to keep track of the chosen convention when composing elements. What truly matters is that the G -action is compatible with the K -action. In the simplest situation one requires

$$\Phi_g(p \cdot k) = \Phi_g(p) \cdot k,$$

so that G commutes with the structure group. More generally one may allow a homomorphism $\psi_g : K \rightarrow K$ and impose

$$\Phi_g(p \cdot k) = \Phi_g(p) \cdot \psi_g(k),$$

in which case G acts by bundle automorphisms twisting the fibres by ψ_g . In either case the induced action on $\text{Fr}(P)$ is well defined and respects the chosen left/right convention.

4.5. G -Invariant Connection 1-Forms and Invariant Ehresmann Connections

We now generalise the definitions of invariance.

DEFINITION 4.8.

A connection 1-form on the principal K -bundle P is a $\mathfrak{k}$ -valued 1-form $\omega \in \Omega^1(P; \mathfrak{k})$ satisfying the reproduction and K -equivariance properties:

$$\omega(\tilde{X}_P) = \tilde{X}, \quad R_k^* \omega = \text{Ad}_{k^{-1}} \omega \quad \forall \tilde{X} \in \mathfrak{k}, k \in K.$$

DEFINITION 4.9. Suppose G acts on P by bundle automorphisms Φ_g . A connection 1-form $\omega \in \Omega^1(P; \mathfrak{k})$ on P is G -invariant if for every $g \in G$

$$\Phi_g^* \omega = \omega.$$

(If Φ_g twists the K -action by ψ_g , the invariance condition should be replaced by the natural twisted equivariance $\Phi_g^* \omega = \text{Ad}_{\psi_g^{-1}} \omega$.)

DEFINITION 4.10. An Ehresmann connection is a smooth horizontal distribution $H \subset TP$ complementary to the vertical distribution $V = \ker d\pi$ and K -equivariant: $(dR_k)_p(H_p) = H_{p \cdot k}$ for all $k \in K$.

DEFINITION 4.11. An Ehresmann connection H is G -**invariant** if for every $g \in G$ and every $p \in P$

$$(d\Phi_g)_p(H_p) = H_{\Phi_g(p)}.$$

(Replace Φ_g by R_g or L_g according to your chosen convention.)

The correspondence between connection 1-forms and Ehresmann connections is standard; the same argument as in the Hopf case shows that this correspondence respects G -invariance.

THEOREM 4.12. *There is a one-to-one correspondence between G -invariant connection 1-forms on P and G -invariant Ehresmann connections on P .*

Proof. The classical map $\omega \mapsto H = \ker \omega$ sends a connection form to a horizontal distribution. If ω is G -invariant then for any $X \in H_p$ and any $g \in G$ we have

$$\omega_{\Phi_g(p)}((d\Phi_g)_p X) = (\Phi_g^* \omega)_p(X) = \omega_p(X) = 0,$$

so $(d\Phi_g)_p X \in H_{\Phi_g(p)}$ and H is G -invariant. Conversely, given a G -invariant horizontal distribution H , the vertical projection (identifying vertical vectors with $\mathfrak{k}$ via the fundamental fields) yields a unique connection form ω with $\ker \omega = H$. The G -invariance of H implies precisely that $\Phi_g^* \omega = \omega$ (or the corresponding twisted formula in the presence of ψ_g). Uniqueness in both directions is immediate from the definitions. $\square$

4.6. G -Invariant Curvature 2-Forms

Before making any definitions, we have this result.

PROPOSITION 4.13. *Let $f: M \rightarrow N$ be a smooth map and let $\alpha \in \Omega^p(N)$, $\beta \in \Omega^q(N)$. We prove that*

$$f^*(\alpha \wedge \beta) = f^* \alpha \wedge f^* \beta.$$

Proof. Fix $x \in M$ and tangent vectors $v_1, \dots, v_{p+q} \in T_x M$. By the definition of pull-back,

$$(f^*(\alpha \wedge \beta))_x(v_1, \dots, v_{p+q}) = (\alpha \wedge \beta)_{f(x)}(df_x(v_1), \dots, df_x(v_{p+q})).$$

Recall the wedge product is defined by the alternating sum

$$\begin{aligned} (\alpha \wedge \beta)_{f(x)}(w_1, \dots, w_{p+q}) &= \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \text{sgn}(\sigma) \alpha_{f(x)}(w_{\sigma(1)}, \dots, w_{\sigma(p)}) \\ &\quad \cdot \beta_{f(x)}(w_{\sigma(p+1)}, \dots, w_{\sigma(p+q)}). \end{aligned}$$

Taking $w_i = df_x(v_i)$ and substituting into the previous display gives

$$\begin{aligned} (f^*(\alpha \wedge \beta))_x(v_1, \dots, v_{p+q}) &= \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \text{sgn}(\sigma) \alpha_{f(x)}(df_x(v_{\sigma(1)}), \dots, df_x(v_{\sigma(p)})) \\ &\quad \cdot \beta_{f(x)}(df_x(v_{\sigma(p+1)}), \dots, df_x(v_{\sigma(p+q)})). \end{aligned}$$

By the definition of pull-back of forms, each factor equals the corresponding pull-back evaluation:

$$\alpha_{f(x)}(df_x(\dots)) = (f^* \alpha)_x(\dots), \quad \beta_{f(x)}(df_x(\dots)) = (f^* \beta)_x(\dots).$$

Thus the right-hand side is exactly the expression for $(f^* \alpha \wedge f^* \beta)_x(v_1, \dots, v_{p+q})$. Since x and the vectors were arbitrary, we conclude

$$f^*(\alpha \wedge \beta) = f^* \alpha \wedge f^* \beta,$$

as required. $\square$

Given a connection ω on P its curvature is the $\mathfrak{k}$ -valued 2-form $F \in \Omega^2(P; \mathfrak{k})$

$$F = d\omega + \frac{1}{2}[\omega \wedge \omega].$$

If ω is G -invariant then F is automatically G -invariant because pull-back commutes with d and with wedge/lie bracket operations:

$$\begin{aligned} \Phi_g^* F &= d(\Phi_g^* \omega) + \frac{1}{2}[\Phi_g^* \omega \wedge \Phi_g^* \omega] \\ &= d\omega + \frac{1}{2}[\omega \wedge \omega] \\ &= F. \end{aligned}$$

Geometrically, this means that the behaviour of parallel transport around small loops does not change as we move along G -orbits.

4.7. Parallel transport and covariant derivatives

In the common special case that G acts trivially on the base points of γ (for instance when G acts only in the fibres or the action on the base is by isometries leaving γ invariant), this reduces to the commutation relation

$$\Phi_g \circ \text{PT}_\gamma^{p_0} = \text{PT}_\gamma^{\Phi_g(p_0)},$$

i.e. parallel transport along a fixed base curve commutes with the G -action.

On any associated vector bundle $E = P \times_\rho V$ (for a representation $\rho : K \rightarrow GL(V)$) a G -invariant connection on P induces a covariant derivative ∇ on sections of E . The G -invariance lifts to the property

$$\Phi_g^*(\nabla_X s) = \nabla_{\Phi_g^* X}(\Phi_g^* s)$$

for all vector fields X on M , sections s of E , and $g \in G$ (again with the appropriate twist if ψ_g appears). This formula precisely captures the statement that covariant differentiation is equivariant with respect to the G -symmetry.

5. Beyond principal bundles: the case of general fibre bundles

Up to this point, our discussion has focused on principal bundles $P \rightarrow M$ with structure group G , where the notions of connection 1-form and curvature 2-form are canonical and well established. A natural question arises: to what extent does the assumption that P is a principal bundle matter? Could one instead begin with a more general fibre bundle $\pi : E \rightarrow M$ with fibre F , and then study the induced geometry on its frame bundle $\text{Fr}(E)$?

Connections on fibre bundles. For a general fibre bundle E , one does not have a group action on the fibre built into the definition. Thus, there is no connection 1-form in the sense of a $\mathfrak{g}$ -valued differential form. However, one can still define an *Ehresmann connection* as a choice of horizontal distribution

$$TE = \mathcal{H} \oplus \ker(d\pi),$$

which provides a notion of parallel transport in E along curves in M . This definition requires only the bundle projection $\pi : E \rightarrow M$, not a principal structure. In this sense, many geometric constructions survive.

Role of the frame bundle. The key point is that every fibre bundle $E \rightarrow M$ has an associated *frame bundle* $\text{Fr}(E)$, whose fibres consist of linear frames of the tangent spaces along the fibres of E . This $\text{Fr}(E)$ is a principal bundle with structure group $GL(\dim F, \mathbb{R})$. Hence, while E itself may not be a principal bundle, $\text{Fr}(E)$ restores a principal setting. Consequently:

- A connection on E can be encoded as a principal connection on $\text{Fr}(E)$.
- The curvature of such a connection is defined in the usual way via the structure equation on $\text{Fr}(E)$.
- Invariance questions (e.g. under an external symmetry group G) can then be formulated directly at the level of $\text{Fr}(E)$.

Comparison with the principal case. Thus, the restriction to principal bundles is not essential for developing a theory of invariant connections and curvature: any fibre bundle E can be studied via its frame bundle $\text{Fr}(E)$, which is automatically principal. The difference is mainly one of perspective:

- In the principal case, the connection form and curvature are built in from the start.
- In the general fibre bundle case, one must first “upgrade” to the principal frame bundle $\text{Fr}(E)$ in order to recover the full algebraic machinery of connection forms and curvature equations.

Geometric intuition. Geometrically, what this means is that while E itself may not carry a natural group action on the fibre, the frame bundle $\text{Fr}(E)$ encodes all of the symmetries of the fibres in terms of linear isomorphisms, and hence admits a canonical principal structure. Connections and curvature live most naturally on this level, and all the invariance arguments developed earlier extend without change once one passes from E to $\text{Fr}(E)$. This shows that the principal bundle framework is not a restriction, but rather the most natural setting in which to phrase the geometry of connections and curvature.

6. Group Actions, Frame Bundles, and Invariant Connections

In this section we investigate what it means for a Lie group action on a manifold to be compatible with the natural geometric structures on its frame bundle. This provides a natural generalisation of the Hopf fibration case: rather than focusing on a specific principal bundle, we consider the frame bundle of an arbitrary manifold equipped with a group action.

6.1. Group actions on the frame bundle

Let M be an n -dimensional smooth manifold, and let G be a Lie group acting smoothly on M :

$$\Phi : G \times M \rightarrow M, \quad (g, x) \mapsto g \cdot x.$$

The frame bundle $\text{Fr}(M)$ is the principal $GL(n, \mathbb{R})$ -bundle whose fibre over $x \in M$ consists of ordered bases (frames) of $T_x M$. The action of G on M lifts naturally to an action on $\text{Fr}(M)$ by pushforward of frames:

$$\tilde{\Phi}_g(x, (e_1, \dots, e_n)) = (g \cdot x, (d(g)_x e_1, \dots, d(g)_x e_n)).$$

This lift is smooth and satisfies the following properties:

- (1) it covers the action on the base, $\pi \circ \tilde{\Phi}_g = \Phi_g \circ \pi$;
- (2) it commutes with the right $GL(n)$ -action, i.e.

$$\tilde{\Phi}_g(u \cdot A) = \tilde{\Phi}_g(u) \cdot A, \quad A \in GL(n, \mathbb{R}).$$

Thus we say that the frame bundle $\text{Fr}(M)$ is G -invariant.

$$\begin{array}{ccc} \text{Fr}(M) & \xrightarrow{\tilde{\Phi}_g} & \text{Fr}(M) \\ \pi \downarrow & & \downarrow \pi \\ M & \xrightarrow{\Phi_g} & M \end{array}$$

6.2. G -invariant connections and curvature

Let $p : \text{Fr}(M) \rightarrow M$ denote the frame bundle with right action R_A of $A \in GL(n)$. Denote the Lie algebra by $\mathfrak{gl}(n)$.

DEFINITION 6.1 Ehresmann connection. *An Ehresmann connection on $\text{Fr}(M)$ is a smooth distribution $H \subset T\text{Fr}(M)$ such that*

$$T_u \text{Fr}(M) = H_u \oplus V_u, \quad V_u = \ker(dp_u),$$

for each $u \in \text{Fr}(M)$, with H right $GL(n)$ -equivariant. We say that H is G -invariant if

$$(d\tilde{\Phi}_g)_u(H_u) = H_{\tilde{\Phi}_g(u)} \quad \forall g \in G, u \in \text{Fr}(M).$$

DEFINITION 6.2. *We say that ω is G -invariant if*

$$\tilde{\Phi}_g^* \omega = \omega \quad \forall g \in G.$$

PROPOSITION 6.3. *There is a bijection between G -invariant connection 1-forms ω on $\text{Fr}(M)$ and G -invariant Ehresmann connections H . The correspondence is given by $H = \ker \omega$.*

Proof. If ω is a G -invariant connection form, then $H = \ker \omega$ is a horizontal distribution. For $X \in H_u$, we compute

$$\omega_{\tilde{\Phi}_g(u)}((d\tilde{\Phi}_g)_u X) = (\tilde{\Phi}_g^* \omega)_u(X) = \omega_u(X) = 0.$$

Thus $(d\tilde{\Phi}_g)_u X \in H_{\tilde{\Phi}_g(u)}$, showing H is G -invariant.

Conversely, given a G -invariant horizontal distribution H , define ω by projecting a vector into its vertical component and identifying this with $\mathfrak{gl}(n)$. This yields a unique connection form with $\ker \omega = H$. Since H is G -invariant, one checks that $\tilde{\Phi}_g^* \omega = \omega$. $\square$

DEFINITION 6.4 Curvature. *The curvature of a connection ω is the $\mathfrak{gl}(n)$ -valued 2-form*

$$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega].$$

If ω is G -invariant, then so is Ω :

$$\begin{aligned} \tilde{\Phi}_g^* \Omega &= d(\tilde{\Phi}_g^* \omega) + \frac{1}{2}[\tilde{\Phi}_g^* \omega \wedge \tilde{\Phi}_g^* \omega] \\ &= d\omega + \frac{1}{2}[\omega \wedge \omega] \\ &= \Omega. \end{aligned}$$

6.3. Parallel transport and the covariant derivative

A connection specifies horizontal lifts of curves γ in M to curves $\tilde{\gamma}$ in $\text{Fr}(M)$. Parallel transport is then given by following these horizontal lifts.

If H is G -invariant, then for any $g \in G$, the image $\tilde{\Phi}_g \circ \tilde{\gamma}$ is a horizontal lift of $g \cdot \gamma$. Thus parallel transport commutes with the G -action. Equivalently, the associated covariant derivative ∇ on any associated bundle satisfies

$$\tilde{\Phi}_g^*(\nabla_X s) = \nabla_{\tilde{\Phi}_g^* X}(\tilde{\Phi}_g^* s).$$

This formalises the idea that differentiation and parallel transport are compatible with the symmetry G .

REMARK 6.

- If G acts by isometries with respect to a Riemannian metric on M , then the Levi–Civita connection is G –invariant.
- When G is abelian, the invariance condition simplifies, and curvature reduces to $\Omega = d\omega$.
- Invariance expresses the idea that parallel transport and covariant differentiation commute with the action of G .

7. Conclusion

This paper has explored how group actions interact with connections and curvature on frame bundles. Starting from the Hopf fibration, we developed the notion of G –invariant connections and curvature forms, and showed in general that invariant Ehresmann connections correspond exactly to invariant connection 1–forms, with curvature inheriting invariance automatically. Geometrically, this ensures that parallel transport and covariant differentiation are consistent with the symmetry of the manifold. Importantly, such invariance is not just a structural property: it provides a practical simplification. On manifolds with symmetry, the conditions of invariance restrict the possible form of a connection and often make it possible to write explicit expressions, particularly in the case of symmetric surfaces.

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References

1. HAMILTON, MARK JD, ‘Mathematical gauge theory’, *Springer* (2017) .
2. Kobayashi, Shoshichi and Katsumi Nomizu. *Foundations of Differential Geometry. Vol. I.* Interscience Publishers, a division of John Wiley & Sons, New York-London, 1963.
3. Bleecker, David. *Gauge Theory and Variational Principles.* Addison-Wesley, Reading, Massachusetts, 1981.
4. Naber, Gregory L. *Topology, Geometry and Gauge Fields: Foundations.* 2nd edition. Texts in Applied Mathematics, vol. 25. Springer, New York, 2011.